

green HOME



A Refined Book on Environmental Education

Name

Class Section



5

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(M.Sc.)

PREPARATORY CURRICULUM
Inculcating 21st-Century Skills

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PREFACE

The **Green Home** is a comprehensive series of Environmental Studies for Classes 1 to 5, based on the latest syllabus of **New Education Policy (NEP) 2020**, and has been prepared according to the guidelines of the **National Curriculum Framework (NCF)**.

The nature of children is very curious. They want to know the answers of all the questions which arise in their minds. The answers of such questions are given in this series. Simple student-friendly language, colourful and well labelled illustrations will help in the teaching of chapters. Picture based activities and a vast variety of exercises will help practising a lot.

As the world is progressing in its environmental problems, such as climate change, global warming, deforestation and pollution, the need of environmental education has become quite necessary. This series enables the students of primary school to develop a solid foundation in this subject, so that they could gain a lifelong love for their planet, Earth and grow to be individuals who are concerned for the environment.

The main features of this series are as follows:

- Powerful visual appeal with multi-coloured pictures.
- Enjoyable skill-based learning.
- Interactive and child centered approach has been followed to arouse interest in the students, to enhance their level of curiosity.

We hope that parents and teachers will give prudential consideration and realistic approach to the book. Any suggestions for the improvement for this book shall be gratefully acknowledged and incorporated, in the subsequent editions.

—Author and Publisher



INCORPORATES NEP 2020

This series has been created with new dimension of joyful learning, along with completely mapped parameters of National Education Policy, 2020.





INSIDE THE BOOK

Outline includes the topics to be learnt in the chapter.

- Food Groups
- Meal
- Water
- Good Food Habits

Recollect includes summary of the chapter.



- People do different kinds of works.
- For everybody, work is an important part of life.
- People do the work to earn money.
- Work done to earn money is called a job or an occupation.

Skill Based Learning stimulates a smart learning for the topics being covered in the chapter which in turn improves the communication and logical skills.



- Art Integration**
- Make a poster on the topic 'water is precious'. Make it colourful and write a message on it about how to conserve water.
 - Present a short skit showing how water gets polluted and what are the do's and don'ts that we should follow to prevent water pollution?



Have a Look! includes some facts related to the chapter to enhance the knowledge of students.



Have a Look!

Plants purify the air by taking in dirty air (carbon dioxide) and giving out clean air (oxygen).

Let's Do! contains carefully graded exercises to test the knowledge of concepts learnt in a chapter.



Let's Do!



A. Tick (✓) the correct option.

- Which part of the plant absorbs water and minerals from the soil?
a. Roots ☐ b. Leaves ☐ c. Stem ☐
- Fruits are:
a. tasty ☐ b. healthy ☐ c. Both a and b ☐

Rack Your Brain includes the questions based on creative thinking.

RACK YOUR BRAIN.

Creative Thinking

- Rajni drank water from a roadside tap, and soon after that he felt sick and started vomiting. Guess the reason.

Teacher's Note provides important information and suggestions on creative approaches to a chapter or a topic.

TEACHER'S NOTE

- Discuss ways of saving water. Tell the students different ways such as using the water that was used to wash vegetables, to water the plants.

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Food

THE FOOD WE EAT

- Components of Food
- Deficiency Disease
- Balanced Diet

One of the most important things in our life is the food we eat. All the living things depend on the environment for food. Our body and mind develop according to the food that we eat. By eating nutritious food, we will grow into healthy adults.

Components of Food

There are certain substances and minerals found in the food that keep our body healthy. They are called **nutrients**.

Nutrients help us to grow and provide the body with the energy it needs for everything it does, from repairing damaged cells to sleeping. Hence, it is important that we understand the role of nutrients in our food.

The nutrients that our body needs are—proteins, carbohydrates, fats, minerals and vitamins. Other than these main nutrients, our body also needs some amount of roughage and water to function properly.

Protein

Our body needs protein to grow taller and stronger. It also helps in repairing the damaged parts of the body. The cells of our hair, nails, muscles and blood contain large amount of protein. Some protein—rich foods include milk, eggs, cheese, curd, fish, chicken, dal, nuts and seeds. Proteins are also known as **body-building foods**.



Carbohydrates

Foods that release the energy quickly are called **carbohydrates**. For doing the smallest action to the largest, our body needs energy. In fact, our body uses energy all the time. Whether we are sleeping or awake, we breathe all the time. All these activities need energy and carbohydrates give the body most of the energy it needs. Some sources of carbohydrates are— wheat, rice, corn, fruits, potato, sugar and honey. They give us energy to walk, study, talk, shout, work, run and play.



Have a Look!

Some foods contain a substance called cholesterol. Cholesterol is an important substance needed by the body. However, when the body has too much cholesterol, it begins to build up in the walls of the blood vessels and blood cannot flow freely through them. This may lead to a heart attack.



Fats

Fats are also energy-giving foods. The body uses fats to maintain its temperature. Fat provides the body with high energy. When we eat more carbohydrates and fats than we need for our daily work and play, the extra food gets stored as fat in our body. Fried and starchy foods, cakes, chocolates, ice creams are the main sources of fats in our diet.

Vitamins and Minerals

Vitamins and minerals can also be classified as **protective foods**. They keep us healthy and found in fruits and vegetables. Vitamins are necessary to fight diseases. They are required in very small amounts.



Minerals are different from vitamins. They serve a variety of purposes. Calcium builds the bones and teeth and helps in blood clotting. Other minerals important for the human body include iodine, magnesium, sodium, potassium and zinc. They are also required in very small amounts for proper functioning of the body.

Roughage

Roughage cannot be digested by the human digestive system. It retains water and improves the working of the intestine by adding bulk to the food. Whole grain bread, green leafy vegetables and some fruits are rich in roughage.



Water

Water is important in many chemical reactions in the body. In fact, every cell in the body must be bathed in water. Water also helps the body get rid of wastes in the form of urine and sweat. It helps to maintain the body temperature. Water makes up about 60 per cent of an adult's body weight.



Have a Look!

Although water has no nutrients, it is essential for life. About 70-80 per cent of the human body is composed of water.

Balanced Diet

To stay healthy, we need all the nutrients in the right proportion. Different foods contain different nutrients. A balanced diet is one that contains all the nutrients in adequate amounts. By having a balanced diet, we will stay fit and healthy.



Deficiency Diseases

Deficiency disease is a disease that occurs due to the deficiency of a nutrient in the diet.

- **Kwashiorkor** and **marasmus** are the deficiency diseases caused among children due to lack of **protein**.
- Lack of **carbohydrates** may result in undernourishments. This leads to improper growth of the body and general weakness.
- Extra **fat** in our diet gets deposited in our body under the skin, under the eyes and in our arteries. This, like cholesterol, prevents the flow of blood in the arteries, causing the heart to pump harder. This could lead to high **blood pressure** and **heart attacks**.



- Lack of **vitamin A** causes night blindness.
 - Lack of **vitamin B** causes beri-beri.
 - Lack of vitamin C and vitamin D causes scurvy and rickets.
 - Lack of iron and calcium causes anaemia and makes our bones weak and soft.
- We should eat protein, carbohydrates and small amounts of fat in our food daily. They give us vitamins and minerals which keep us healthy and protect us from **germs** or **microbes**.

Deficiency diseases cannot spread from one person to another.

How are hunger and disease related?

Many diseases are caused when people do not get enough food to eat or do not eat the right kind of food. **Malnutrition** is a condition that occurs when a person does not get food to eat.

It is important to think before eating, and it is very important to eat the right kind and quality of food to stay healthy.



- ♦ All the living things depend on the environment for food.
- ♦ It is important to understand the role of nutrients in our body.
- ♦ Proteins are also known as body-building foods.
- ♦ A balanced diet contains the right amounts of carbohydrates, proteins, fats, minerals, and vitamins that our body needs.
- ♦ Diseases caused by the lack or shortage of nutrients are called deficiency diseases.



A. Tick (✓) the correct option.

1. Lack of iron causes anaemia / scurvy.
2. Roughage can / cannot be digested by human digestive system.
3. Minerals are found in fruits / pulses.
4. Fats are body-building / energy-giving foods.
5. To stay healthy, we need all the nutrients in the right / wrong amount.

B. Match the following.

Column (A)	Column (B)
1. Beri-Beri	a. Vitamin - D
2. Scurvy	b. Iron
3. Anaemia	c. Vitamin - B
4. Rickets	d. Vitamin - C
5. Marasmus	e. Protein

C. Put the following foods into correct columns.

butter, bread,	ghee, rice,	oil, vegetables,	fruits, chicken,	dal, cheese	milk,
Carbohydrates	Fats	Proteins			

D. Answer the following questions.

1. What are nutrients? What are their main types?

2. What is a balanced diet?

3. What are proteins? Give some examples of food items rich in proteins.

4. Why should fat be eaten in moderation?

Skill Based Learning



A. Solve these riddles.

This is a vegetable that grows underground. It makes you cry when you cut it with a knife.



This fruit is sweet or sour. It grows in bunches on vines.



Life Skills and Values

- B.** Children in poor families do not eat a balanced diet, because of which they remain malnourished. Spread awareness among the poor by telling them about the importance of having nutritious food.

RACK YOUR BRAIN.

Creative Thinking

- What would happen if we ate only one kind of food everyday?

TEACHER'S NOTE

Explain the importance of vitamins and minerals to the students in detail.



BE HEALTHY, BE SAFE

- Diseases Caused by Germs
- Preventing Communicable Diseases

- How Communicable Diseases Spread?

A balanced diet and regular exercise help us to be healthy. Still we may sometimes fall sick. This is caused by germs which spread through air, water, food or contact with an infected person.

Diseases Caused by Germs

Germs are very tiny organisms that cannot be seen by the naked eye. They are visible only under a microscope. Protozoa, bacteria, virus and fungi are some such germs.

- **Protozoa** cause diseases, like malaria, amoebiasis, and kala-azar.



- **Bacteria** cause diseases, like pneumonia, typhoid, tuberculosis, cholera and diarrhoea.

- **Viruses** cause diseases like common cold, flu, chickenpox, polio and AIDS.



- Fungi mostly cause infections of skin, eyes and hair.

How Communicable Diseases Spread?

A disease that spreads from an infected person to another person is called an **infectious** or a **communicable disease**.

Germes are so tiny that we can only see them with the help of a microscope, but they are there in the air, unclean water and food. When we breathe them in or drink or eat the infected food, we can get sick.

Lets see how these germs are spread.

Through Air

Microbes which cause diseases, like flu, cold and tuberculosis (TB), get into the air when the infected person sneezes or coughs. It may also be caused by breathing in the same air due to being in the same room for a long time. When some healthy person touches the infected things, he can get infected by the germs. This may spread the germs through air or direct contact.



Have a Look !

Many good microbes are also found in the surroundings. Curd is full of good microbes.

Through Contaminated Food and Water

Microbes which cause cholera, jaundice, polio, typhoid and diarrhoea, are often spread from infected food and contaminated water.

People who have microbes in their bodies pass some of them in their stool. Stools passed in the open can easily get into water bodies.

Through Insects and Animals

Diseases like malaria, dengue and plague spread through mosquitoes, flies, etc. If

female anopheles mosquito bites a healthy person, the germs get transferred into that person's blood and cause malaria.

- Plague is transmitted by rats. If not controlled, it may result in an epidemic.



Preventing Communicable Diseases

We know that infections spread mainly from touch, unclean water and food, air and through insects. There are many ways of preventing communicable diseases.

Let us see how:

- Keep yourself clean. Bath regularly. Wash your hands with soap and water before having the food and after going to the toilets. Maintain proper sanitary conditions.
- Keep your surroundings clean. Throw garbage into a well-covered dustbin.
- Drink clean, boiled, filtered water, because water can be made safe for drinking by these methods.
- Wash your vegetables and fruits well before you cook or eat them.
- Food must be covered to prevent flies sitting on it. Milk is heated or pasteurised to prevent harmful bacteria from growing in it.
- To prevent certain diseases, ask your parents to take you for vaccination at regular intervals.

When we get vaccinated or injected by special vaccines, our body builds up its own immunity. Vaccines are available for diseases like polio, cholera, typhoid, jaundice, measles, meningitis etc.

Get enough rest and sleep well. It is good for you! A healthy diet, regular exercise, proper sleep and clean surroundings can save us from getting sick with either deficiency or communicable diseases.



- ♦ Our body needs a balanced diet to keep healthy.
- ♦ Protozoa, bacteria, virus, and fungi are some germs which cause diseases.
- ♦ Communicable diseases spread through air contact, food, water and insects.
- ♦ A healthy body helps us to work well and be happy.



A. Tick (✓) the correct option.

1. Cough is caused by:
 a. virus ☐ b. protozoa ☐ c. bacteria ☐
2. Germs can be seen by a:
 a. naked eye ☐ b. telescope ☐ c. microscope ☐
3. Bacteria cause a disease like:
 a. cholera ☐ b. amoebiasis ☐ c. flu ☐
4. Infections spread mainly from:
 a. touch ☐ b. clean water ☐ c. vaccines ☐

B. Given below are names of certain diseases.

common cold, malaria, plague, cholera, jaundice, diarrhoea

Try to find out their mode of transmission and fill in the following table.

Diseases spreading through air	Diseases spreading through food or water	Diseases spreading through animals

C. Write 'T' for true and 'F' for false.

1. Anaemia makes you active and fit.
2. All diseases are caused by virus.
3. Deficiency diseases are caused by more vitamins and minerals in our food.
4. Our body needs a balanced diet to keep healthy.

D. Answer the following questions.

1. Mention four ways by which germs can spread.

2. What are the ways to prevent communicable diseases?

3. What are infectious diseases?

4. Define germs.

Skill Based Learning

A. When did you fall Sick last time? Did you have any fever, vomiting or pain because of your illness? Did the other members of your family fall sick around the same time with the same illness? Why did it happen? Discuss.

Life Skills and Values

B. On your birthday, visit an orphanage and arrange a meal for the children living there. You may also buy some gifts, sweets, chocolates, biscuits, etc. and distribute them among the children. The cheer on the faces of the children will make your birthday memorable.

RACK YOUR BRAIN.

Creative Thinking

- What do you think is the main cause of pneumonia? Justify your answer.

TEACHER'S NOTE

Explain how to prevent communicable diseases to the students in detail.



The World of Living

REPRODUCTION IN PLANTS

- Reproduction in Plants
- What is Seed Dispersal?

Like animals and humans, plants are living things, and they also have the ability to reproduce and increase their numbers.

Reproduction in Plants

Plants reproduce in different ways. They reproduce through:

- Seeds
- Roots
- Spores
- Stems
- Leaves

Plants Reproduce from Seeds

Most flowering plants reproduce from seeds. Once flowers bloom they turn brown, the petals fall off and they change into fruits. Fruits have seeds inside them. New plants grow from these seeds.

Seed is the reproductive part of a plant. Plants produce many seeds, because all the seeds do not grow into new plants due to various reasons, like:

- Some seeds are not healthy and are destroyed by winds or heavy rains.
- Some seeds are eaten by animals and some do not get the right conditions to germinate.

Germination

Seeds only germinate when they get the suitable conditions of air, water, light and

warmth. If these conditions are fulfilled, a seed grows into a new plant called **seedling**. This process of a seed growing into a plant is called **germination**.



Seed leaves are also called **cotyledons**. They contain food for the baby plant or embryo. The baby plant uses his food till it grows its own roots and leaves. Once the food in cotyledons is used, the cotyledons shrink and fall off. Seeds like pea and bean contain two cotyledons and are called **dicotyledons**.

After the food from cotyledons is used, young plants continue making their own food through **photosynthesis**.

Plants Reproduce From Stems

Some plants like rose, sugarcane and money plants can reproduce by stem cuttings. The stem cuttings develop the roots when put in moist soil.

Some underground stems like onion, potato and ginger have buds on them and any piece of the vegetable having bud can grow into new plants.



Plants Reproduce from Roots

Plants like sweet potato, radish, carrot and turnip grow new plants from roots.

If we cut the top portion of a carrot or radish and plant it into the ground, new plant will grow from it.



Plants Reproduce from Leaves

Plants like bryophyllum produce new plant from the margins of its leaves. The tiny plantlets fall off in the ground and grow as an independent plant.

Plant Reproduce from Spores

Plants like ferns and mushroom produce from spores which grow into new plants. On falling on the moist soil, the spores are found on the underside of the leaves.



What is Seed Dispersal?

Seeds need air, water and sunlight to germinate. **Dispersal** is the scattering of seeds far away from the parent plant. Seeds have special features which help them to disperse. Also there are some other factors which help them to disperse. These factors can be wind, water, animals, etc. These are called **dispersal agents**.

Agents of Dispersal

Wind

Some seeds are very small and light. The seeds of orchid, hiptage and drumstick are

almost as fine as dust. These are carried to a new place by the wind.

Seeds like dandelion and cotton have hairy growths which carry the seeds far away from the parent plant.



Water

Like lotus, plants that grow near the water have fruits and seeds with spongy or fibrous outer covering which helps them to float away from the mother plant.

Coconut can travel for thousands of kilometres across seas and oceans without decaying. When it reaches the shore it germinates, if it gets the right conditions.

Animals

Some plants have juicy fruits that animals and birds like to eat. The seeds of these fruits pass into their excretion which are left in different places. This can be far away from the parent plant.

Cherry, jamun and apple seeds are dispersed in this way. When the seeds get the right conditions they germinate.



Have a Look!

Squirrels collect nuts like acorns and bury them for winter food, but they often forget where they have buried them and these grow into new trees.

Explosion

Some fruits like the ladyfinger, peas and bean explode or burst open with great force, scattering the seeds away from the mother plants.



Recollect .o

- ♦ Plants reproduce through seeds, stems, roots, leaves and spores.
- ♦ A seed needs air, warmth, water and soil to germinate.
- ♦ Seed is a reproductive part of a plant.
- ♦ When a new plant grows from a seed, it is called germination.
- ♦ Wind, water, animals and explosion are the agents of dispersal.



A. Tick (✓) the correct option.

1. Moss / carrot reproduce from spores.
2. The seeds of bean / cotton are dispersed by explosion.
3. The cotyledons / seed coat provide food to the embryo.
4. Plants like ferns / lotus produce from spores.

B. Give an example for each of the following.

1. Plants that grow from stems _____
2. Plants that grow from spores _____
3. Plants that grow from leaves _____
4. Plants that grow from seeds _____
5. Plants that grow from roots _____

C. Which part of the following plants will grow into a new plant?

- | | | | |
|--------------|-------|----------|-------|
| Sugarcane | _____ | Rose | _____ |
| Potato | _____ | Fern | _____ |
| Sweet potato | _____ | Carrot | _____ |
| Mushroom | _____ | Turmeric | _____ |

D. Answer the following questions.

1. Define germination. What are the essential conditions for germination?

2. What are the special features of the seeds which get dispersed by winds?

3. What is dispersal? Why is it important?

4. How are animals helpful in the dispersal of seeds?

Skill Based Learning

A. Can you name a fruit whose seeds you eat?

B. Can you remove the seed coat of a seed? If yes how?

Life Skills and Values

C. Plants keep our environment clean & healthy. We should plant more and more trees to maintain the balance in nature and increase the green cover around us.

RACK YOUR BRAIN.

Creative Thinking

- What would happen if all the seeds of a plant grew in the soil under the mother plant?

TEACHER'S NOTE

Explain the concept of reproduction in plants to the students in detail.

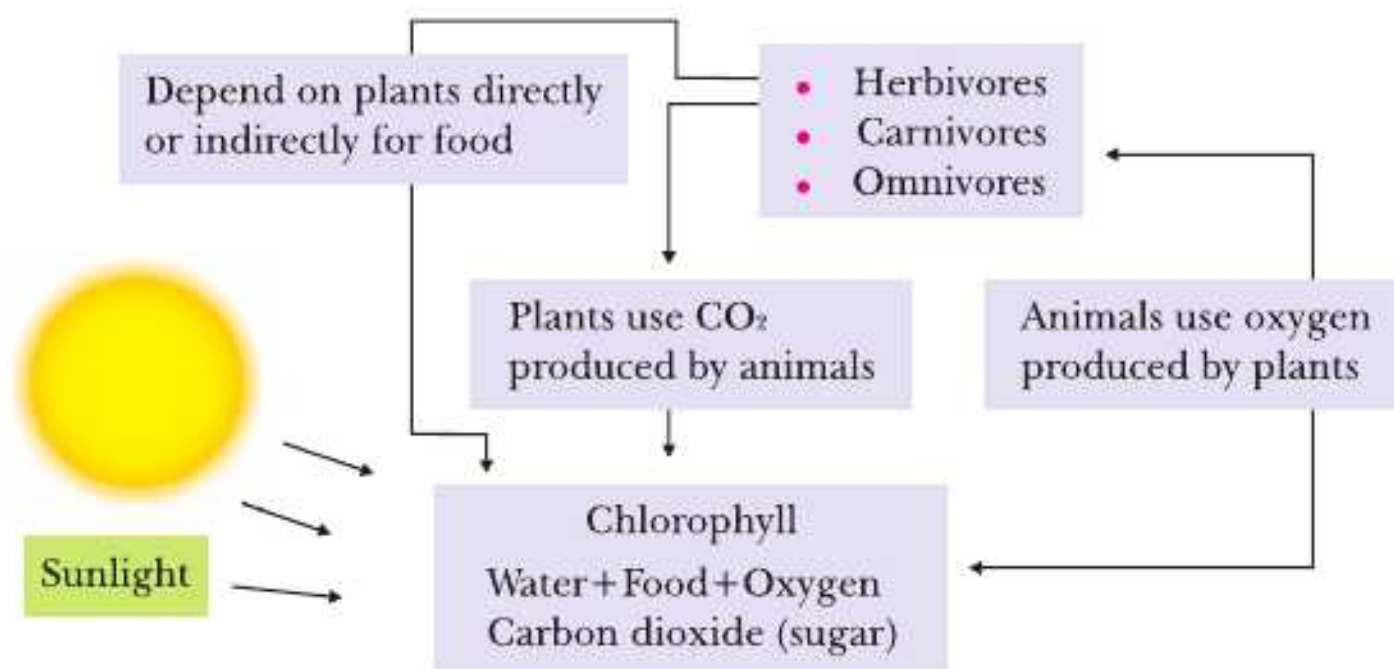


PLANTS AND ANIMALS : LIVING AND SURVIVING

- How Do Plants Depend on Animals?
- Food Chain

- How Do Animals Depend on Plants?

Being living things, plants and animals depend on each other and also share a lot of similarities like breathing, reproduction, etc.

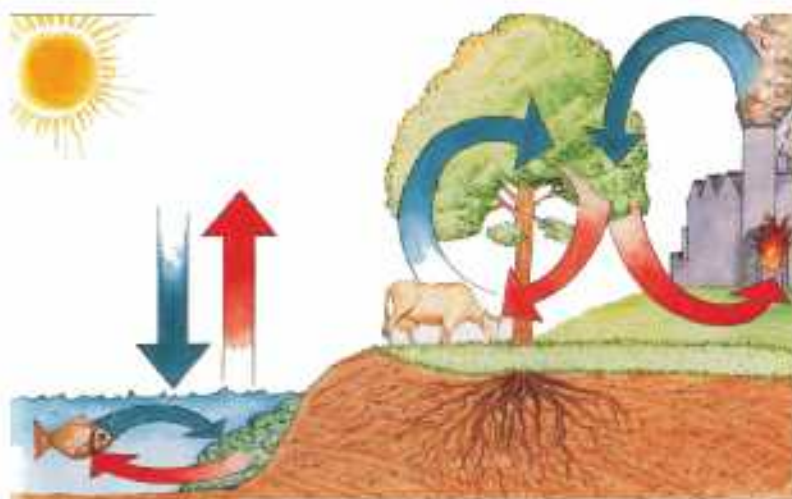


Animals live in large numbers where there are plants. This is because animals depend on plants for their survival.

How do Plants Depend on Animals?

Plants Depend on Animals for Carbon Dioxide

Plants take in carbon dioxide for photosynthesis and give out oxygen, thus providing oxygen for all living things to breathe.



Plants Depend on Animals to Disperse Seeds

Some seeds have hair on them while some have hooks and others have sharp spines. These stick to the bodies of animals and are carried far away from the place of their origin, thus animals help in the dispersal of seeds.

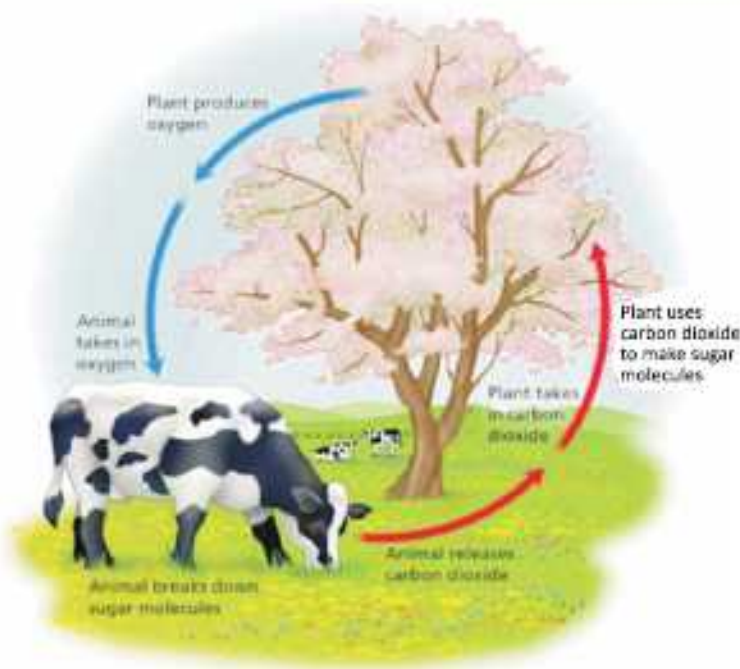


Plants Depend on Animals to Produce Seeds

Many kinds of flowering plants depend on insects, like bees, moths and butterflies to produce fruits with healthy seeds.



How Do Animals Depend on Plants?



Animals Depend on Plants for Oxygen

Plants make their food through a process called **photosynthesis**. The oxygen produced by the plants during photosynthesis mixes with the air. Animals and plants use this oxygen to live. All animals depend on plants for a constant supply of oxygen.

Animals Depend on Plants for Shelter

Many kinds of animals live on trees and other plants. Plants are home for a large variety of big and small animals, like monkeys, leopards, bats, birds, ants, squirrels, etc. When a tree is cut, many animals become homeless.

BE A FRIEND TO TREE



Animals Depend on Plants for Food

Many kinds of animals eat only plants. Such animals are called **herbivores**. All herbivore animals do not eat the same kind of plants.

For example : The bats eat fruits and the langur, monkeys eat tender leaves and fruits of plants. Cows and buffaloes eat grass and herbs.

Carnivores like snakes, tigers, owls, eagles eat only animals. They depend indirectly on plants for food by eating herbivores.

Omnivores eat both plants and animals. Bears, foxes, peacocks, ants and sparrows are omnivores.

Food Chain

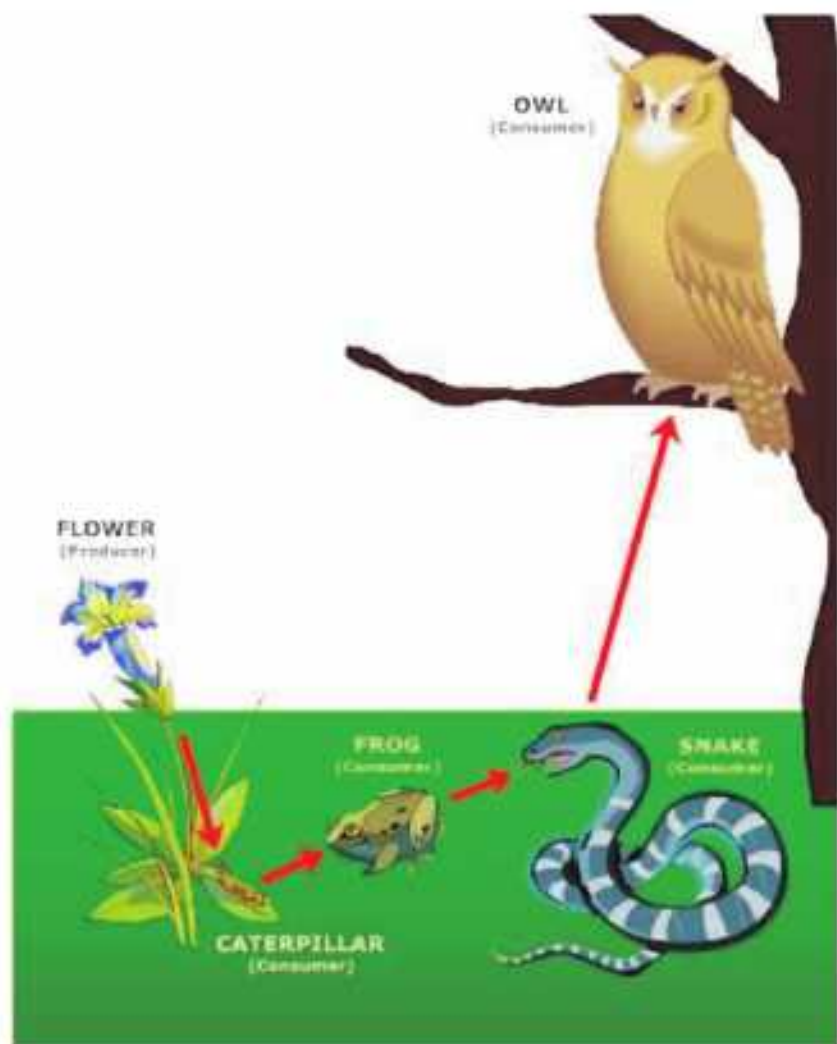
We already know that green plants prepare their own food. That is why they are called **producers**. Plants use sunlight, water and carbon dioxide to produce food.

Animals, on the other hand, consume plants or animals as food. Animals are called **consumers**.

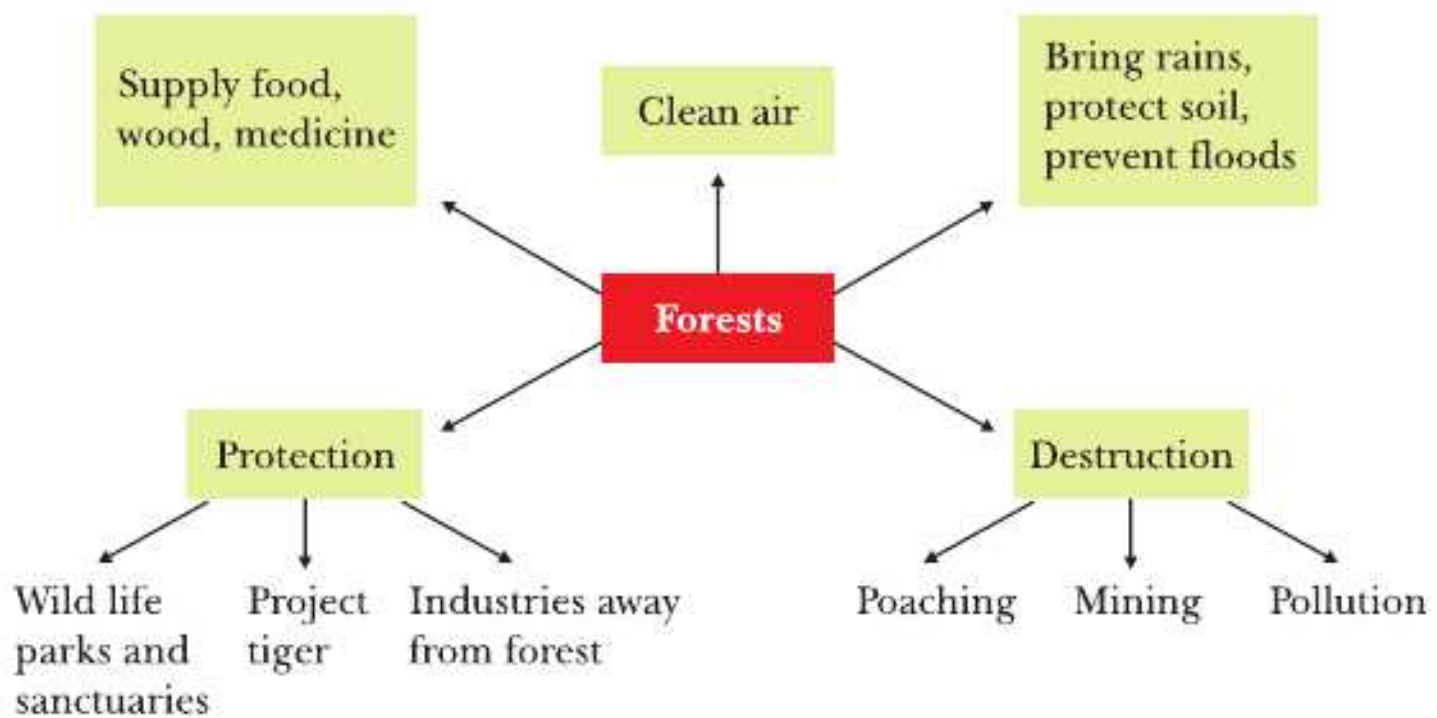
A food chain is a **sequence of living things** found in a region, in which each living thing is **food for the next member** in the sequence.

The plants are eaten by the herbivores and the herbivores are eaten by the carnivores. When the carnivores die, the scavengers feed on them. The uneaten parts are then decomposed by the decomposers. The decomposers absorb the nutrients and pass them to the soil. These nutrients are again absorbed by the roots of the plants to prepare food.

This chain is called a **food chain**. The food chain shows the interdependence between the living things in the environment.



Forest is a Home to Many Kinds of Animals



A forest not only has trees but also a rich growth of shrubs, herbs, creepers and climbers. Forests are very useful to us.

Forests Clean Air

We get oxygen from plants in a forest.

Forests Provide Medicines

The leaves, roots and bark of many forest plants are used to make medicines.

Forests Supply Food

Forests provide many kinds of edible fruits.

Forests Supply Wood

Forest supply timber which is used to build houses, furniture and ships.

Forests Prevent Floods

The dense growth of plants block the flow of rainwater.

Forests Protect the Soil

Plants prevent soil from being washed away by flowing rainwater and wind.

How Can We Protect Forests?

To protect a forest, both its plants and animals must be protected.

- Trees should not be cut in large numbers.

- Roads must not be constructed through a forest.
- People should not purchase clothes, shoes, bags or any other articles made from fur or bones of wild animals.
- Industries that pollute air or water should not be allowed to be set up close to a forest.
- Some people kill wild animals for their meat and skin or horns, which is called poaching. Hunting and poaching should be strictly prohibited.

Recollect

- ◆ Plants depend on animals for carbon dioxide.
- ◆ Some animals help in seed dispersal.
- ◆ Plants are called producers and animals are called consumers.
- ◆ Animals depend on plants for oxygen, shelter and food.
- ◆ We get oxygen from plants in a forest.



A. Tick (✓) the correct option.

- Animals depend on plants for:

a. oxygen	☐	b. food	☐	c. Both a and b	☐
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- Herbivores, carnivores and omnivores are collectively called:

a. producers	☐	b. consumers	☐	c. omnivores	☐
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- Roads must not be constructed through:

a. plants	☐	b. forest	☐	c. sunlight	☐
-----------	--------------------------	-----------	--------------------------	-------------	--------------------------
- Plants make their food through a process called:

a. photosynthesis	☐	b. carbon dioxide	☐	c. sunlight	☐
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B. Read the sentences carefully and rewrite them correctly.

- Omnivores eat only plants. _____
- Plants are called consumers. _____
- Carnivores do not depend on plants for food. _____
- We get carbon dioxide from plants. _____
- To protect a forest, only animals need to be protected. _____

C. Draw labelled diagrams to describe the following.

1. Food chain
2. How plants and animals depend on each other for oxygen and carbon dioxide?

D. Answer the following questions.

1. How do animals depend on plants?

2. What is a food chain?

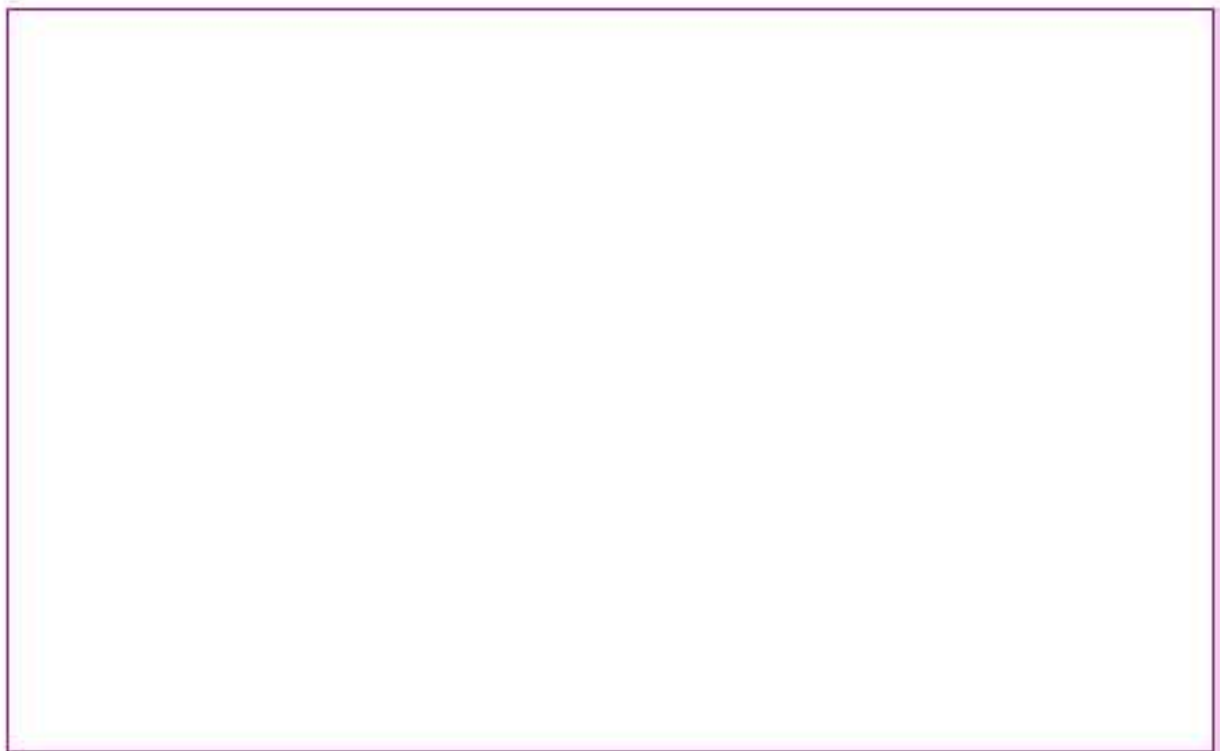
3. List the uses of forests.

4. How do animals help in seed dispersal?

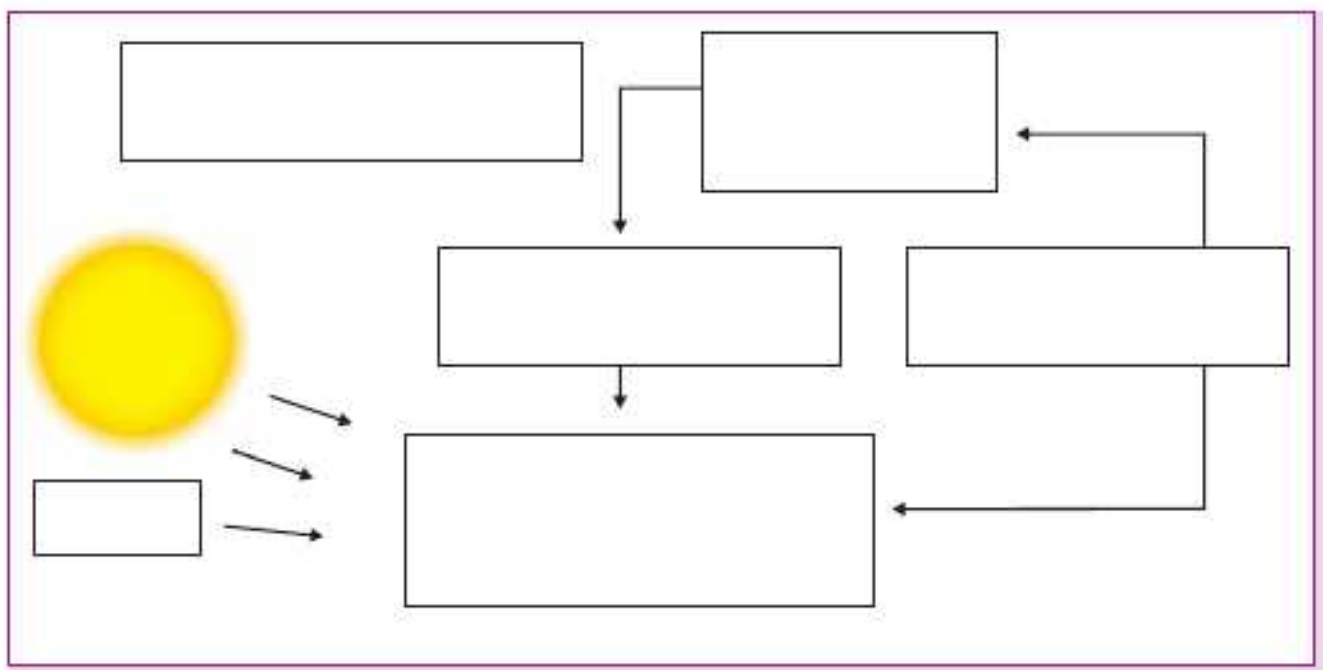
5. Write the names of animals that are found in the forest of our state.

Skill Based Learning

A. Draw the trees and the animals you observed in and around the trees.



B. Complete this chart that shows the interdependence of plants and animals.



Life Skills and Values

- C. Forest dwellers in India are the poorest and most backward communities. They are often asked to leave their homes to make way for mining and logging companies. There are many NGOs that work for the betterment of these communities. We must extend our support to such NGOs.

RACK YOUR BRAIN.○

Creative Thinking

- If forests are community resources and tribals are allowed to use them, then why are timber companies not allowed to cut trees and use these resources?

TEACHER'S NOTE

Explain how do plants depend on animals to the students in detail.



INSIDE OUR BODY

- External Organs
- Care of Organs

- Internal Organs

Our body has many organs. An organ is a part of the body that performs a specific and vital function. Organs are of two types: External and Internal.

External Organs

They can be seen and touched. Eyes, ears, hands and legs are some external organs.

Internal Organs

They are present inside our body. We are unable to see the internal organs but they are essential for survival. They are protected by the skeleton. Lungs, heart and brain are some internal organs.

Brain

Brain is the most important organ of the human body. It is **soft** and **delicate** like jelly. It controls the working of all other organs in the body. It is protected within the skull which is made up of several strong and hard bones joined with **immovable** joints.

The human brain is a highly complex and efficient organ. Every second, it sends out billions of messages to every cell of the body, in the form of electrical impulses. It works non-stop. It helps us to remember things. This ability to remember and recollect the information is called **memory**.

The brain has three main parts: cerebrum, cerebellum and medulla.

Cerebrum

The cerebrum is the largest part of the brain and controls the sense organs, memory, basic intelligence and thinking and learning ability. It also controls the functions of the eyes, ears, nose and tongue.

Cerebellum

The cerebellum controls the muscle movement and maintains the balance of the body.

Medulla

The medulla is the lowermost part of the brain. The spinal cord begins at the medulla. It controls breathing, heartbeat and the blood circulation.

Lungs

There are two lungs in the human body, one on either side of the chest. The heart and lungs are protected by a part of the skeleton known as the **rib cage**.

When we breathe in, the air goes in through the nose, down the windpipe (called the trachea) in the lungs.

The lungs are filled with many air sacs called **alveoli**. When the air reaches the **alveoli**, the blood vessels lining the walls of the alveoli take in the oxygen from the air and give out the carbon dioxide that they have brought from the various parts of the body.

This oxygenated blood is taken to the heart, from where it is pumped to every cell of the body. The air in the lungs now is full of carbon dioxide. This is pushed out of the body through the nose. When we breathe in, the lungs expand and when we breathe out, they contract. The muscle below the lungs controls the breathing process by expanding and relaxing. This muscle is called the **diaphragm**.



Heart

The heart is the pump which causes the blood to circulate in the body. It is located on the left side of the chest. The blood not only delivers oxygen to every cell for **respiration**, but also food and water besides any other substance, which requires to be sent from one part of the body to another. Under normal conditions in an adult, the heart beats 72 times per minute.

The human heart is four chambered. It has two **auricles** (also called **atrium**) a right auricle and a left auricle and two **ventricles**, a right ventricle and a left ventricle. The right side of the heart receives deoxygenated blood (which contains carbon dioxide to be thrown out of the body) from various parts of the body by means of veins. The left side of the heart receives oxygenated blood (full of oxygen) from the lungs. The oxygenated blood is then carried to various parts of the body by **arteries**. The **capillaries** link the arteries and the veins. Arteries, veins and capillaries together constitute the **blood vessels**.

Care of Organs

Be Good to Your Brain

- Wear a helmet when you ride a bicycle or ride on a motor vehicle.
- Play outdoors everyday.
- Solve puzzles, read books and play music to give exercise to your brain.
- Reduce watching television or playing video games.
- Sleep for at least 7 to 8 hours. Your brain needs rest.

Recollect

- ♦ Our body has many organs.
- ♦ External organs can be seen and touched.
- ♦ Lungs, brain and heart are some internal organs.
- ♦ Human brain is a highly complex and efficient organ.
- ♦ The human heart has four chambers.



A. Tick (✓) the correct option.

1. The lowermost part of the brain is the:
a. cerebrum ☐ b. medulla ☐ c. cerebellum ☐
2. The _____ link the arteries and the veins.
a. blood vessels ☐ b. capillaries ☐ c. auricles ☐

3. Human heart has two _____.
- a. veins ☐ b. auricles ☐ c. arteries ☐
4. The cerebellum controls the movement of:
- a. muscle ☐ b. skeleton ☐ c. digestion ☐

B. Given below are some organs of the human body.

eyes	nose	kidney	ears
lungs	heart	hand	tongue

Classify them as external and internal organs in the table below.

External organs	Internal organs
_____	_____
_____	_____
_____	_____
_____	_____

C. Write short notes on the following.

- Human brain
- Alveoli
- Rib cage

D. Answer the following questions.

- Discuss the functions of the brain and the heart.

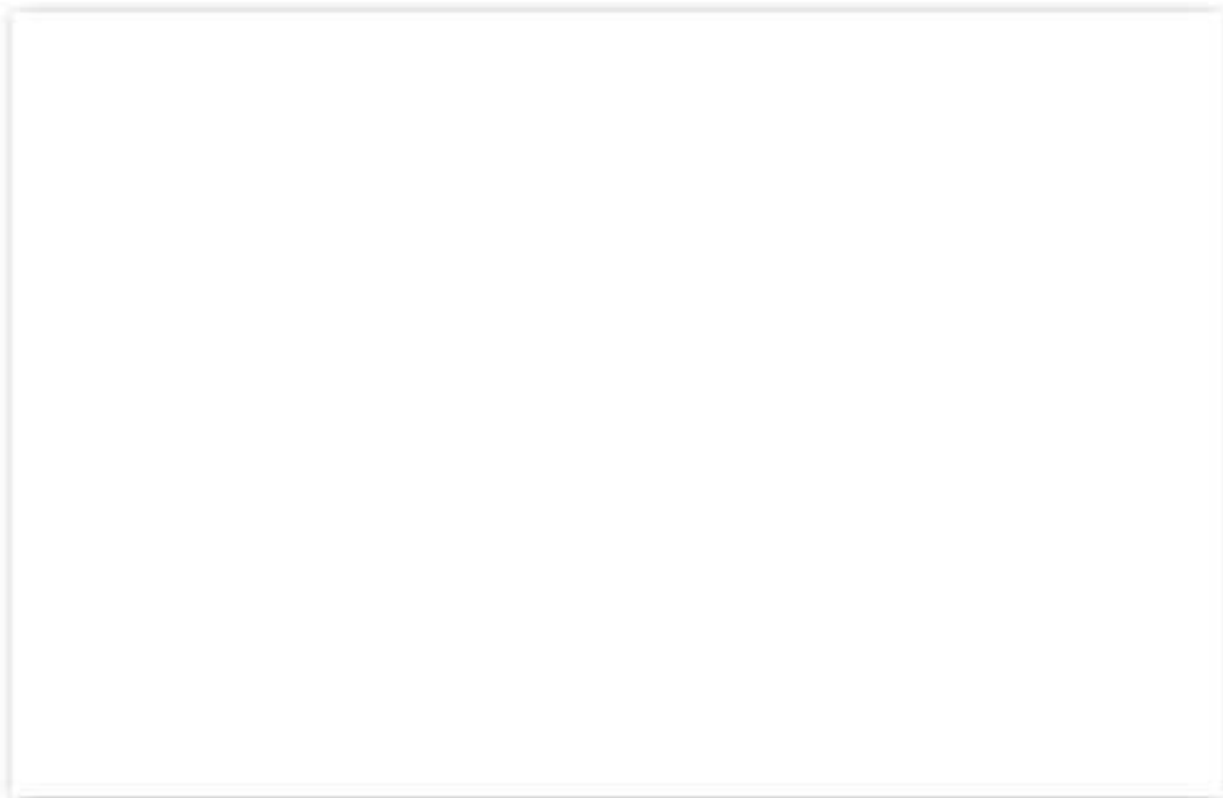
- What are internal organs?

- How can you take care of your brain?

- What is the function of lungs?

Skill Based Learning

- A. Draw an outline of the human body. Now draw all the internal organs that you have studied in the chapter. Also label all the organs.



Life Skills and Values

- B. Sometimes when we go near a slum area, we smell a stink. To escape it, we cover our nose. But imagine how the people living in that area must be feeling. So next time when you pass a slum, do not cover your nose, instead make a group and go to the slum, spread awareness about cleanliness and help the people clean their area.

RACK YOUR BRAIN.

Creative Thinking

- A grown up offers you a chocolate for not telling anyone, that he touched you. What will be your reaction to such a situation.

TEACHER'S NOTE

Explain how should we take care of organs to the students in detail.



Play, Event and Monuments

GAMES AND SPORTS

- Different Types of Games
- National Team

- Team Spirit
- Eminent Sport Personalities

A **game** is an activity which provides pleasure and excitement and develops our physical, mental and social abilities. They are undertaken usually for enjoyment and relaxation. A game in which a person or a team competes against another person or team, is called a **sport**.

Different types of Games

Games and sports have always been an important pastime for adults and children alike. Many of the games involve physical exercise, which keep us healthy. We learn important skills, such as communication and cooperation quite easily through games. We learn to share, take the responsibility and abide by rules through our games.

- Chess, carrom, ludo and playing cards are some of the popular **indoor games**.
- Other games that need a large area to play are played outdoors. Cricket, football, hockey, basketball, etc. are popular **outdoor games**.



There are many games that are played by a single person. These games are known as **individual games**. Some of them are swimming, skating and cycling. Here a player has to achieve success for himself or herself.



Games like cricket, football, hockey, etc. are considered as **team games** because they are played between two teams. Each team has a definite number of players. These games are played according to a set of rules. In team games, there is an umpire or a referee who supervise that both the teams follow the rules and regulations.

Team Spirit

In a team game, every player plays according to the need of the team on the basis of understanding, cooperation, coordination and respect among the team members. It is called team spirit. Team spirit creates selflessness and happiness among players. A good team is one in which the members feels free to contribute their individual ideas.

National Team

A national team is the team that is selected to play a sport for the country. It consists of players from all parts of the country.

When our national teams go out to play for our country, they have only one identity—**Indian**.

Players in a national team usually belong to different regions, religions and economic backgrounds. They play together to represent their country, and their main objective is to win honour for their nation.

When our national team wins a tournament, the whole country celebrates its victory. The national game of our country is field hockey. So far, India has won eight Olympic gold medals in hockey. Cricket is a popular sport in India.



Eminent Sport Personalities

Every person whether a boy or a girl has a right to sound health and physical fitness. Girls and boys both are encouraged to develop their skill in different types of games and sports.

Sachin Tendulkar

Sachin Tendulkar or 'Master Blaster' as he is popularly known as, is a batting sensation. He has crossed many milestones. He became the first cricketer in the world to score more than 12,000 runs in one-day cricket.



Mahendra Singh Dhoni

Mahendra Singh Dhoni did the miracle by winning first T20 world cup in 2007 and cricket world cup in 2011.

Anju Bobby George

Anju Bobby George has won many championships in woman athletics.



Sania Mirza in tennis, Saina Nehwal in badminton, Anjali Bhagwat in pistol event and Joshna Chinappa in squash have made a great name for themselves.



Abhinav Bindra

He won the individual Olympic gold medal in 108m Air Rifle shooting event at the Beijing Olympic games, 2008.



Rajyavardhan Singh Rathore

He is an ex-Indian Army officer who won a silver medal for India in shooting at the Athens Olympics, 2004.

Neeraj Chopra

He is the first Asian athlete to win an Olympic gold medal in Javelin (2020 Tokyo) and the first Asian to win gold in his event at the world championship (2023 Budapest).



Recollect

- ♦ A game is an activity undertaken usually for relaxation and enjoyment.
- ♦ Different types of games are: indoor games, outdoor games, individual games and team games.
- ♦ Team spirit creates selflessness and happiness among players.
- ♦ Each team has a definite number of players.



A. Tick (✓) the correct option.

1. A _____ is an activity which provides pleasure and excitement.

a. game

☐

b. worship

☐

c. reading

☐

2. Indoor games are played in the:

a. room

☐

b. ground

☐

c. road

☐

3. Game played by the single person is known as:
 a. indoor game ☐ b. outdoor game ☐ c. individual game ☐
4. Popular sport in India is:
 a. hockey ☐ b. cricket ☐ c. swimming ☐

B. Fill in the blanks.

1. _____ has won a great name for herself in athletics.
2. Each team has a _____ of players.
3. We learn important skills such as _____ and _____ quite easily through games.
4. In cricket, there are _____ umpires on the field.
5. _____ is the national game of India.

C. Write 'T' for true and 'F' for false.

1. Chess, carrom and ludo are outdoor games.
2. Many of the games involve physical exercise, which keep us healthy.
3. There are nine players in a cricket team.
4. Captain is the head of the team.
5. Sachin Tendulkar is also called Master Blaster.

D. Answer the following questions.

1. What is the difference between games and sports?

2. Differentiate between individual games and team games?

3. Define team spirit.

4. Do boys and girls play together?

Skill Based Learning

- A. Can you name some more sports persons who have been cricket captains of India?
- B. Find out the names of some more sportswomen in India.

Art Integration

- C. Draw the pictures of two of your favourite indoor and outdoor games and also write their names.



Life Skills and Values

- D. We should be careful while playing in a park. We should not run over the tiny saplings there or hit someone with the ball.

RACK YOUR BRAIN.○

Creative Thinking

- If you lose a game, should you congratulate your opponent on their success or not? Give reasons to support your answer.

TEACHER'S NOTE

Explain the importance of games and sports to the students in the class in detail.



SOME SPECIAL CELEBRATIONS

- National Days
- International Days

In the calendar, some days are marked as special or memorable days. These days may be national days and international days.

- National days are celebrated within the country, like Gandhi Jayanti, Teacher's Day, etc.
- International days are celebrated throughout the world, like world environment day, earth day, etc.

National Days

Independence Day

On 15th August 1947, India got its independence from British rule. This important event is celebrated every year on the same day by the entire nation. We get together at school, sing patriotic songs and remember the martyrs. On this occasion, the Prime Minister of India hoists the national flag at the Red Fort in Delhi and delivers a speech.



The National Flag is also unfurled in every school and government institution. The National Anthem is sung with great pride. Sweets are distributed and kites are flown to

celebrate this occasion. Every Indian remembers the sacrifices made by the freedom fighters and salutes the National Flag.



Have a Look!

Jana-Gana-Mana is our National Anthem. It was originally composed in Bengali by Rabindranath Tagore.

Republic Day

It is celebrated on 26th January each year. On 26th January 1950, India became a republic. A republic is a country in which the head of the state or the President is elected by the people.

On this day, our President hoists the national flag and take salute of the parade. A big parade is organised at Rajpath, in Delhi. Important units of the Army, some selected schools and cultural troupes from almost all states of India walk in an orderly procession along the Rajpath. While the units of our Defence forces, the Air force, the **Navy** and the **Army** present their best tanks, aircrafts and other machines, the school children and state troupes present cultural programmes. The state units make models of best scenes of their state, for example, a harvest scene. These are mounted on trucks and shown to the people and called **Tableaux**. On this day, prizes are distributed by the president to selected children for their acts of bravery. Thousands of young and old people alike come to witness the parade. The Prime Minister, all ministers and even foreign ambassadors and guests attend the parade show. Similar celebrations are also held in the state capitals.

The governor of the concerned state takes the salute.



Gandhi Jayanti

Mahatma Gandhi was born on 2nd October 1869. His birthday is celebrated as Gandhi Jayanti. Mahatma Gandhi led us when we were fighting for independence from British rule. He is remembered as the **Father of the Nation**. He is affectionately called **Bapu**. On his birthday, the President, the

Prime Minister, many leaders and people visit Rajghat in Delhi on the bank of river Yamuna. They pay their floral tribute to the Father of the Nation.

Teacher's Day

We celebrate Dr S. Radhakrishnan's birthday as Teacher's Day. It is celebrated on 5 September. Dr S. Radhakrishnan was the second President of India. He was a great scholar, teacher and a philosopher. We celebrate this day to honour our teachers. Our teachers make sincere efforts to teach, train and groom us. They also help us become better human beings and responsible citizens of the country. On this day, students perform stage programmes for teachers. They also thank their teachers for teaching them.



International Days

World Environment Day

It is celebrated every year on 5 June. It was declared by the United Nations General Assembly in 1972. Every year on this day, celebrations are hosted by a different environment related theme.

World Health Day

On 7 April every year, World Health Day is celebrated by the member countries of the World Health Organisations (WHO). The main objective is to make people aware of the importance of good health. On this day, many health camps are organised. Doctors conduct free medical check-ups. They also give health tips.



Earth Day

Earth Day is celebrated on 22 April every year. It was started in 1970 by Gaylord Nelson.

On Earth Day, many cleanliness campaigns are organised all over the world. This day is celebrated to promote a healthy environment.



- ♦ National days are celebrated within the country.
- ♦ International days are celebrated throughout the world.
- ♦ On Independence Day, the Prime Minister of India hoists the National Flag at the Red Fort in Delhi.
- ♦ Earth Day is celebrated on 22 April every year.
- ♦ Doctors conduct free medical check-up on World Health Day.



A. Tick (✓) the correct option.

1. National festivals are celebrated by all throughout the:

a. religion ☐	b. country ☐	c. region ☐
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2. The President distribute bravery awards to the children on:

a. Gandhi Jayanti ☐	b. Republic Day ☐	c. Teacher's Day ☐
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3. The Prime Minister of India hoists the National Flag at Red Fort on:

a. New Year ☐	b. Independence Day ☐
c. Gandhi Jayanti ☐	
4. He is remembered as the Father of the Nation.

a. Mahatma Gandhi ☐	b. Dr S. Radhakrishnan ☐
c. Gaylord Nelson ☐	

B. Given below are pictures of some great leaders of our country. Can you name them?







C. Write 'T' for true and 'F' for false.

1. India became a republic in 1950.
2. World Health Day was started by Gaylord Nelson.
3. Dr S. Radhakrishnan's samadhi is at Raj Ghat.
4. Pandit Jawaharlal Nehru is called the Father of the Nation.

D. Answer the following questions.

1. Why is Teacher's Day celebrated?

2. How is Republic Day celebrated?

3. When and how is the World Health Day celebrated?

4. When is Mahatma Gandhi's birthday celebrated?

Skill Based Learning

A. List the names of activities that are required to keep our earth clean. Then discuss about them with your friends.

Life Skills and Values

B. On special days, we should pledge to put smiles on the faces of poor and underprivileged children. We may distribute sweets and some other gifts to such children.

RACK YOUR BRAIN.

Creative Thinking

- With Whom do people exchange gifts and greet each other happily.

TEACHER'S NOTE

Explain the importance of national days to the students in detail.



OUR HISTORICAL MONUMENTS

- Historical Monuments
- To Preserve Our Historical Buildings

India is a land of monuments. Historical buildings and monuments are silent witness of our magnificent and historical past and cultural heritage. These are places of historic, scientific and scenic interest. Our glorious past and rich valued heritage is reflected in our majestic monuments. The prosperous economy of India has always attracted the foreigners.

Historical Monuments

When invaders came to India, they brought many new ideas of architecture, painting, sculpture and other crafts with them. Our historical buildings are a unique blend of the influence of European and Iranian styles and our own native style.

The architecture of the monuments and their paintings and carvings tell us about the people at that time and their social, political and religious beliefs.

Let us look at some monuments of India.

Taj Mahal

The Taj Mahal is one of the seven modern wonders of the world. It is situated in the city of Agra about 200 kilometres from Delhi, on the banks of the river Yamuna. It was built by the Mughal Emperor Shah Jahan for his beloved wife, Mumtaz Mahal. The Taj Mahal took about 20 years and 20,000 men to



complete. It was constructed in Persian, Indian and Islamic style and designed by Persian architect. The majestic white marble has given colour and beauty to the monument.

Qutub Minar

One of the finest monuments in India, the Qutub Minar was started by Qutu-bud-din Aibak in 1193 and completed by his successor Iltutmish. It is 239 ft high and the highest stone tower in India.



Ajanta and Ellora Caves

The Ajanta and Ellora caves are known for cave temples and architecture that attract the tourists in large number. These are the best examples of rock—cut architecture, which was built by the Pallavas who ruled there. Both the caves have an interesting mix of scriptures from Hindu, Buddhist and Jain faiths. These are situated at Aurangabad.

The Sanchi Stupas

Among the earliest monuments that have survived, are the Buddhist stupas. These are funeral monks of Buddhist monk. The smooth round shape of the stupa symbolises the perfection and immortality. The Sanchi Stupa was built in Madhya Pradesh by Emperor Ashoka in the third century BC.



Jama Masjid

One of the largest mosques in India, Jama Masjid was built by Shah Jahan in Delhi. More than 5000 workers toiled for six years to complete this mosque.

To Preserve Our Historical Buildings

Historical buildings and monuments are silent witnesses of our glorious past. They narrate stories of the past and tell us about the bravery of Indian rulers. These buildings have been declared as public property by the government and are known as **Heritage Buildings**.

How Can We Protect Our Historical Monuments?

There are numerous historical buildings in India. While visiting these monuments, we must:

- Cooperate with the caretakers of the buildings.
- Observe general rules of cleanliness in the public places.
- Inform the staff on duty if somebody tries to harm them.



- ♦ India has a very rich historical past and cultural heritage.
- ♦ We would do our best to protect these monuments.
- ♦ Historical monuments are silent witnesses of our glorious past.



A. Tick (✓) the correct option.

- Shah Jahan built the Taj Mahal in the memory of his:
a. daughter ☐ b. son ☐ c. wife ☐
- The Taj Mahal was built on the banks of river:
a. Ganga ☐ b. Narmada ☐ c. Yamuna ☐
- Ajanta and Ellora caves are situated at:
a. Mumbai ☐ b. Aurangabad ☐ c. Ghaziabad ☐
- Historical buildings are witnesses of our glorious:
a. past ☐ b. present ☐ c. future ☐

B. Use each clue to guess the name of the historical monument and write it in the given space.

- This stupa was built by emperor Ashoka. _____
- One of the seven wonders of the world. _____

3. This monument is 239 ft high. _____
4. These caves are situated at Aurangabad. _____

C. Find out where these monuments are located.

1. Taj Mahal _____
2. Ajanta and Ellora caves _____
3. The Sanchi Stupa _____
4. Jama Masjid _____

D. Answer the following questions.

1. What attracted the foreign rulers to invade India?

2. What do you know about the Taj Mahal?

3. Why should we protect our monuments?

4. What can we do to protect our monuments?

5. What is stupa? What does it consist of?

Skill Based Learning



- A. Find out the names of some more buildings constructed by the Mughals.**

- B. Describe about any one monument that you have visited.**

Life Skills and Values

- C. Make a poster to bring awareness about conservation of our historical monuments.**

RACK YOUR BRAIN.

Creative Thinking

- Why should we plant trees around our historical monuments?

TEACHER'S NOTE

Explain the importance of historical monuments to the students in detail.



Safety

FIRST AID

First Aid

First Aid

First aid is the aid or treatment given to the injured before the arrival of a doctor. First aid may also be described as the medical assistance given to an injured or sick person for immediate relief.

In spite of being careful, sometimes accidents do happen. Accidents can hurt or injure us for a long time. In case of any accident, you should first inform an adult or a doctor.

Fracture

When a bone breaks or cracks, it is called a fracture. This can cause unbearable pain. Do not press the area of fracture and avoid moving it. It has to be treated immediately. Send for a doctor.

The fractured area needs support. Gently place a newspaper, magazine or cardboard on fractured area for the support and tie it up. If it is a hand, do not let it hang down. The support you have given to the fractured area is called a sling. A sling is a support made from cloth. In an emergency, you can even use a **duppatta** for sling.



Burns

Burns can be classified in three types.

Simple Burns

It can be treated at home. Cool the burnt area with cold water or ice for 10-15 minutes till the pain becomes less. Apply antiseptic lotion.

Severe Burns

It can form the blisters. Never touch or prick the blister, as it will then become an open wound and will catch infection. Severe burns should be treated by a doctor.

Chemical Burns

It needs to be treated by a doctor. Before the doctor arrives, wash the affected area with plenty of water and lightly cover it with a piece of clean cloth.

Cut

Cuts often occur while using sharp objects, like knives and blades. Wash the cut with soap and water. Apply an antiseptic lotion on it.



Protect the cut area from dust and germs by putting a band-aid on it. If the cut is deep, take the victim to a doctor as he or she may require stitches.

Bleeding from the Nose

Many people bleed from the nose in summers when small blood vessels (capillaries) burst. To stop bleeding from the nose, pinch the bridge of the nose and apply ice on the head. If ice is not available, pour cold water on the head.

Ask the victim to breathe through the mouth. If the bleeding does not stop, take the patient to the doctor.

Sprain

An injury caused in a joint or muscle is called a **sprain**. Sprain can be very painful. Put the sprained part in cold water on the first day and then in warm water from the second day onwards. For support, tie a bandage in a “figure of eight (8)”. If the pain is unbearable, the patient must be taken to the doctor.

Electric Shocks

Electric shocks could be fatal. In case of an electric shock, first switch off the main power supply. If the person suffers from burns, cover the affected area with a sterile bandage. A clean cloth could also be used. In case the shock is severe, immediately rush the victim to the hospital. You should be very careful while handling electrical gadgets. Don't touch them with wet hands. Also, wear rubber—soled shoes or slippers while handling them.

- ♦ Most accidents happen because of carelessness.
- ♦ In case of a fracture, do not move the person.
- ♦ In case of burns, put the affected area under running water.
- ♦ For nose bleed, apply ice on the head and pinch the bridge of the nose.



A. Tick (✓) the correct option.

1. Treatment given to the injured person is called:

a. first aid ☐	b. support ☐	c. sling support ☐
---------------------------------------	-------------------------------------	---
2. When a bone breaks or cracks it is called:

a. wound ☐	b. fracture ☐	c. electric shock ☐
-----------------------------------	--------------------------------------	--
3. Burns can be divided into:

a. two parts ☐	b. three parts ☐	c. four parts ☐
---------------------------------------	---	--
4. Many people bleed from the nose in:

a. summer ☐	b. winter ☐	c. rainy season ☐
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B. Write 'T' for true and 'F' for false.

1. A sling support the fractured area.
2. In a sprain, the injured area swells up.
3. To put off electrical fire, you should use water.
4. Antiseptic should be applied on the burnt area.

C. Fill in the blanks.

1. Protect the cut area from _____ and _____.
2. In case of an electrical fire, put off the _____.
3. When a bone cracks or breaks, it is called a _____.
4. _____ is an injury at a joint.
5. Electric shocks could be _____.

D. Answer the following questions.

1. Define first aid.

2. What is a sling?

3. Why should we not use water in case of an electric shock?

4. What first aid should be given to a person who has

(i) fractured his leg?

(ii) sprained his ankle?

Skill Based Learning



A. Have you ever got burnt on your hand or leg? How did you treat it?

B. Have you ever given any first aid to someone?

Life Skills and Values

C. If you find someone lying injured at home on the road or in school, you should immediately call an adult so that immediate medical help can be provided.

RACK YOUR BRAIN.🧠

Creative Thinking

- Why should we not play near open ditches or manholes?

TEACHER'S NOTE

Explain the importance of first aid to the students in detail.



Nature

NATURAL CALAMITIES

- Natural calamities
- Flood

- Cyclone
- Fire

- Earthquake

Natural Calamities

A natural calamity is the effect of a natural event that causes great loss to the environment and human life. Natural calamities, such as earthquakes, floods, cyclones may strike at any time.

Cyclone

Cyclones are strong and rotating winds, common in coastal areas, and are associated with heavy rains and high tides, and even floods. Cyclones can be forecasted by warning systems along coastlines. These can reduce the effects caused by it. Cyclone shelters also help to reduce the loss of human life. Bolt plantation helps to check the soil erosion.



Have a Look !

Most earthquakes start 16 km below the surface of the earth.



Earthquake

An earthquake is the sudden shaking of the earth caused by movements under the surface of the earth. Earthquakes occur constantly around the world. Often they are too small for people to feel at all. Earthquakes are measured on the **Richter Scale**. It is believed that

earthquakes that measure 2 or less on the Richter Scale, cannot be felt at all. Earthquakes are very destructive. They can cause great damage to life and property. The **Bhuj** earthquake that occurred on 26 January, 2001 in Gujarat, is the most disastrous earthquake, ever recorded in the history of India. More than 20,000 people died, over 15 lakhs were injured and about 8 lakh houses were damaged.

Floods

Heavy and continuous rainfall makes the rivers overflow which causes floods in nearby areas.

Floods can also be caused by the collapse of a dam, which makes the excess water flow into adjoining areas.

Melting of snow in the summer causes river to overflow. Coastal areas are more prone to the floods, as the rapid change in air pressure may result in sea storms, known as cyclones or typhoons followed with heavy rain and storm. This causes floods in the coastal areas. Floods cause damages to the crops, lives and properties. Along with floods come the most dreaded diseases like malaria, cholera, typhoid and dengue.

Fire

Fire is one of the major causes of accidents. During heavy thunderstorms, if lightning falls on any object, it may catch fire. A fire may also break out because of a leak in the cooking gas pipe, or a short circuit in the electric supply system. Whatever the reason may be, a fire causes extensive damage to life and property. Buildings, such as schools, colleges, offices, cinema halls, shopping malls and large residential complexes should strictly follow fire safety measures.



Some of them are:

- A proper fire exit.
- A fire extinguisher within reach.
- Maintaining fire alarm systems for detecting and warning purposes.
- Conducting fire drills at regular intervals.

- ♦ Earthquakes cause major damages to life and properties.
- ♦ Floods occur due to heavy rainfall.
- ♦ Coastal areas are more prone to floods.
- ♦ Fire is one of the major causes of accidents.
- ♦ Earthquakes occur constantly around the world.



A. Circle the odd one out.

earthquake	snake bit	fracture
flood	water	fire
gas pipe leakage	short circuit	earthquake

B. Fill in the blanks.

1. In the year _____, India had a major earthquake in Gujarat.
2. Earthquake can cause many natural _____.
3. _____ is a scale to measure earthquakes.
4. More than _____ people died in Bhuj earthquake.
5. Earthquake occur _____ around the world.

C. Match the following words with their pictures.

1. Earthquake
2. Flood
3. Cyclone
4. Fire



D. Answer the following questions.

1. How is an earthquake measured?

2. What is a natural calamity?

3. How are floods caused?

4. Which are the dreadful diseases that spread during floods?

Skill Based Learning

A. Have you ever experienced only natural disaster?

B. Have you helped the people affected by a disaster in any way?

Life Skills and Values

C. Disaster management is very important. There are many disaster awareness programmes designed for students. Learn more about such programmes and gather other important informations about disaster management on these websites:

<http://www.ndmindia.nic.in>

<http://geohxz.org>

RACK YOUR BRAIN.

Creative Thinking

- If a building near your house collapsed and some people were trapped in it, how would you help them?

TEACHER'S NOTE

Explain the different natural calamities to the students in detail.



SOILS OF INDIA

- Soil
- Soil Erosion

- Different Types of Soil

Soil

Soil is the **uppermost layer** of the earth. Without soil, no plants can be grown. All living beings directly or indirectly depend on soil. Soil is formed by breaking up of rocks. Rains and flowing water break down the rocks into small pieces. These small particles are carried to different places by running water and wind. It takes millions of years for soil to be formed.

Different Types of Soil

A variety of crops can be grown due to the different types of soils. Soils are of different types and vary from region to region.

Alluvial Soil

Himalayan rivers bring with them large quantities of alluvium. This is deposited on the river banks. It is one of the most fertile soils. These are found in the northern plains, coastal plains and river deltas. A large number of important crops, such as rice, pulses, jute, wheat, sugarcane and vegetables grow in this type of soil.

Black Soil

The north-western part of Deccan plateau has black soil. This soil is made up of the lava of the volcanic rocks. It is clayey and can retain water for a long period. It is good for growing cotton crops, wheat, sugarcane and millets.

Red Soil

The rest of the Deccan plateau has red soil. Red soil is derived from igneous rocks in the hot and relatively dry parts of the southern part and eastern part of Indian Peninsula. The red soil is also called **laterite soil**. This soil is less fertile but with the use of fertilisers, it can yield good crops. It develops reddish colour due to the presence of iron in the rocks. This soil is very good for growing pulses, jowar, bajra, groundnuts, coffee and tea.



Desert Soil

Desert soil is found in the Thar desert of Rajasthan. It contains sand but lacks in humus content. It is good for growing crops like dates, gram, wheat, etc.

Mountain Soil

In the mountains, the soil is very coarse. It consists of pebbles and stones. This soil is suitable for **orchards** where fruits are grown. **Terrace farming** is practised on hills.

Soil Erosion

The upper layer of soil contains a lot of minerals and humus for plants to grow. Manures and fertilisers are added to the soil to make it more fertile. But trees are cut and grass is pulled out, which make the soil infertile. Sometimes rain water and strong winds also take the soil away. This is called **soil erosion**.

- We must take steps to stop soil erosion.
- Trees should not be cut thoughtlessly.
- We should plant more trees. Trees help to hold the soil together.
- More trees should be planted in the areas with heavy rainfall.
- Overgrazing by animals must be controlled.
- Embankments should be built around the fields.



- ♦ Soil is the uppermost layer of the earth's crust.
- ♦ It takes millions of years for the formation of soil.
- ♦ India has many different types of soils.

- ♦ Alluvial soil is the most fertile soil.
- ♦ Cutting of trees and pulling out grass make the soil infertile.



A. Tick (✓) the correct option.

- Terrace farming is used in:

a. desert area ☐	b. hilly area ☐	c. plain area ☐
---	--	--
- Mountain soil is good for:

a. orchards ☐	b. pulses ☐	c. tea and coffee ☐
--------------------------------------	------------------------------------	--
- Other name of red soil is:

a. laterite soil ☐	b. clayey soil ☐	c. black soil ☐
---	---	--
- Desert soil is good for:

a. coconut ☐	b. dates and gram ☐	c. orchards ☐
-------------------------------------	--	--------------------------------------

B. Match the following.

- | | |
|------------------|-------------------------------------|
| 1. Red soil | a. Gujarat |
| 2. Desert soil | b. Himalayan region |
| 3. Black soil | c. Eastern part of Indian Peninsula |
| 4. Mountain soil | d. River delta |
| 5. Alluvial soil | e. Thar desert |
| 6. Laterite soil | f. Chota Nagpur Plateau |

C. Write 'T' for true and 'F' for false.

- Without soil, no plants can be grown.
- Alluvial soil is not very fertile.
- The red soil is also called laterite soil.
- We should not control cutting down of trees.
- The soil in the Deccan plateau is sandy.

D. Answer the following questions.

- Why is soil formation a slow process?

2. Name the different types of soil?

3. What is soil erosion? What are the main causes of it?

4. What type of soil is found in mountains?

5. How can we prevent soil erosion?

Skill Based Learning

A. Plant a tree in your house and take care of it.

Life Skills and Values

B. Collect some pictures depicting soil erosion from books, internet, etc. Now take a sheet of chart paper and make a collage of pictures showing soil erosion.

RACK YOUR BRAIN.

Creative Thinking

- How does overgrazing by domestic animals cause soil erosion?

TEACHER'S NOTE

Explain different types of soil to the students in detail.



NATURAL RESOURCES

- Natural and Man-Made Resources
- Non-Renewable Resources
- Renewable Resources

Natural and Man-Made Resources

Resources are things available in the environment that we need and depend on in order to live. We get food, air and water from nature. Besides these, we also get many things of daily uses like paper, fuel, timber, metals and so on. We derive all these things from trees, rocks and minerals present in the earth. Air, water, soil, sunshine, rocks, minerals, plants and animals are called **natural resources**.

Man-made resources are those products which are the result of activities done by people in the fields of agriculture, housing, transportation and so on.

Now, we shall study about some renewable and non-renewable resources.

Renewable Resources

Renewable resources are those that can be returned to nature if they are not overutilised. Once renewable resources are consumed at a rate that exceeds natural rate of replacement, their quantity will diminish and eventually run out. Renewable natural resources include soil, water, sunlight and wind. However, some renewable resources can exhaust if we use them faster than they are renewed.



Soil

Soil is the topmost layer of the earth's surface. It is made of finely crushed rock. Soil makes it possible for plants to live and grow. All living things, as you know, depend directly or indirectly on plants for food. It is also a home for many animals, like snakes, rabbits and earthworms. Soil is mixed with other things to make useful products, like bricks and cement.



Conservation of Soil

The following steps should be kept in mind to conserve the soil:

- Avoid overgrazing and the overuse of land for crops.
- Keeping your soil healthy by adding plenty of compost.
- Disposing the garbage carefully so that the soil is not destroyed.
- Make sure that the plants grow on a soil rich in humus, because it binds the soil.



Have a Look!

In our country, the Chipko Movement was started to stop the cutting of trees. People saved trees from being cut down by hugging them. This prevented the woodcutters from felling the trees.

Water

Water is another natural resource that we need. Though most of the earth is covered with water, only a small portion of it can be used for drinking and growing plants. This water is called **potable water**. Rainfall is the biggest source of potable water, provided it does not fall through polluted air. Apart from drinking, we also use water to make things, to travel, to make electricity and to wash things.



Conservation of Water

Some ways to save water are:

- Turning off the tap while brushing your teeth.
- Not washing the vegetables and fruits under running water.

- Using a bucket for soapy water for washing cars and other vehicles.
- Rainwater harvesting is a modern method used for conserving and making use of rainwater.

Air

All living things need air to breathe. Animals and plants would die without it. Air also holds aeroplanes and kites above the ground and it is used to inflate tyres and balloons. Air consists of a mixture of gases. Nitrogen and oxygen are the main gases found in the air. Oxygen is also needed for burning fuel and making electricity.



Non-Renewable Resources

Non-renewable resources are natural resources that exist in a fixed amount. They cannot be remade, regrown or regenerated as fast as they are consumed and used up.

Some non-renewable resources can be renewed but take an extremely long time to renew. Fossil fuels, for example, take millions of years to form and so are not considered 'renewable'. Coal, petroleum and natural gas are all fossil fuels.

Conservation of Non-Renewable Resources

The following steps should be followed to conserve non-renewable resources.

- Saving electricity by switching off electrical appliances when not in use.
- Clean fuel like CNG should be used in vehicles.
- Avoiding wastage of cooking gas by employing proper cooking methods.
- Avoiding wastage of minerals.



- Resources are the things available in the environment.
- Resources are of two types: natural and man-made.
- Soil is the topmost layer of the earth's surface.
- All living things need air to breathe.



A. Tick (✓) the correct option.

1. Soil is made up of finely crushed:
a. ice ☐ b. rocks ☐ c. animals ☐
2. The water used for drinking is called:
a. river water ☐ b. potable water ☐ c. pond water ☐
3. Clean fuel like CNG should be used in:
a. Vehicles ☐ b. homes ☐ c. schools ☐
4. The topmost layer of the earth's surface is:
a. soil ☐ b. wind ☐ c. air ☐

B. Match the following.

Natural Resources

Man-made Resources

paper

soil

fuel

water

sunlight

C. Fill in the blanks.

1. _____ are things available in the environment.
2. Renewable natural resources include _____ , _____ , _____ and _____ .
3. _____ is the biggest source of potable water.
4. _____ is also needed for burning fuel and making electricity.

D. Answer the following questions.

1. What are natural resources?

2. How can we conserve the soil?

3. What would happen if there is no air on earth?

4. Name the gases present in the air.

5. Why should water be conserved at all times?

Skill Based Learning



Art Integration

A. Draw and colour a picture of overgrazing by animals in the given space.

B. Draw posters to create awareness about local resources among people.

Life Skills and Values

C. Water is an elixir of life, a very important natural resource. Your EVS teacher wants you to prepare a plan for a formative assessment activity– “How to save water, the vital natural resource.”

RACK YOUR BRAIN.

Creative Thinking

- “Water is important.” Give the reason for this statement.

TEACHER'S NOTE

Discuss the importance of soil and water conservation to the students in detail.



REDUCING POLLUTION

- Pollution
- Different Kinds of Pollution

Pollution

Pollution is the common enemy of most natural resources. Burning dry leaves and smoke from factories pollute the air. Dyes used to colour fabrics, when released into water pollute it. Petrol pumps and garages that dump their waste like grease are polluting the soil. We must, therefore, strictly control the activities that cause pollution.

Different Kinds of Pollution

Pollution can be classified in many ways, like air pollution, water pollution, noise pollution, etc.

Air Pollution

The mixing of harmful gases and other particles in the air is called **air pollution**.

Effects of Air Pollution

Air pollution causes many pollutants to enter the human bloodstream through the nose, mouth, skin and the digestive tract. These pollutants produce harmful effects on the body. They cause skin and respiratory allergies. They also harm our eyesight. Disease like asthma, cancer and tuberculosis are also caused by air pollution.



Causes of Air Pollution

Air pollution is caused by harmful gases, like carbon dioxide, carbon monoxide, nitrous oxide and so on. These are called **air pollutants**.

They are released by:

- burning fossil fuels, like coal and natural gas.
- automobile exhausts.
- air conditioners and refrigerators.

How Can We Control Air Pollution?

- Plant more trees. This will help to increase the oxygen content in the atmosphere.
- Use public transport. Also request your parents to use car pooling. This will reduce traffic on the roads and hence the pollution.
- Encourage factory owners to use filters in chimneys of factories. This will make the smoke free from poisonous substances.

Water Pollution

The mixing of harmful wastes in ground water, lakes, rivers, seas and oceans by human activities is called **water pollution**. It is a major problem faced by most countries.

Effects of Water Pollution

Polluted water if used for irrigation affects the quality of crops. It decreases agriculture production. Poisonous chemical from factories affect, marine life. Plants, fish and living things in polluted water bodies die due to these chemicals. If we eat such fish, we will also fall ill.



Causes of Water Pollution

- Industrial water is a major source of water pollution.
- People pollute water by littering and pouring oils down the sink. They also use fertilisers and other chemicals in their gardens. These pollutants are then washed into drains.
- In villages, bathing cattle, washing utensils and clothes in river water and ponds also add to water pollution.

How Can We Control Water Pollution?

- Recycle and dispose the garbage properly.
- Use the dustbins provided. Keeping beaches free from garbage will go a long way to help keep our water clean.
- Encourage farmers and gardeners to use natural fertilisers like compost or manure. Composting helps to retain soil moisture.

Noise Pollution

The presence of loud noise in the atmosphere is called noise or sound pollution. Any sound louder than 90 DB (decibels in which intensity of sound is measured) is harmful for our ears.

Nowadays, sound or noise pollution has become one of the biggest environment problems.

Effects of Noise Pollution

Overexposure to noise can cause hearing loss. It can also lead to high blood pressure, loss of sleep, increased heartbeat and rate of breathing and so on.

Causes of Noise Pollution

The rising number of vehicles on the roads, busier airports, machines, electrical gadgets, loudspeakers and so on create noise pollution.



Same home appliances, like the mixer-grinders, vacuum cleaners, hair dryers, air conditioners and so on also add to noise pollution.

How Can We Control Noise Pollution?

- Request everyone to use sound absorbers in noisy generators. This will make the generators soundproof.
- Encourage your family members to maintain vehicles and home appliances for smooth functioning.
- Encourage people working in noisy areas to wear earplugs.

- ♦ Pollution is the common enemy of most natural resources.
- ♦ Air pollution cause skin and respiratory allergies.
- ♦ Polluted water if used for irrigation, affect the quality of crops.
- ♦ The presence of loud noise in the atmosphere is called noise or sound pollution.



A. Tick (✓) the correct option.

1. Presence of loud noise creates:

a. air pollution ☐	b. water pollution ☐	c. noise pollution ☐
---	---	---
2. Pollution caused by harmful gases is called:

a. noise pollution ☐	b. air pollution ☐	c. water pollution ☐
---	---	---
3. The mixing of harmful wastes in groundwater causes:

a. water pollution ☐	b. air pollution ☐	c. noise pollution ☐
---	---	---
4. Pollution can be divided in:

a. two parts ☐	b. three parts ☐	c. many parts ☐
---------------------------------------	---	--

B. Match the following.

Water pollution	loud speaker
Air pollution	garbage
Sound pollution	harmful gases
	poisonous chemicals
	industrial waste
	traffic on the road

C. Write 'T' for true and 'F' for false.

1. Oxygen and Nitrogen are harmful gases.
2. Home appliances do not cause noise pollution.
3. Sound absorbers should be fitted in noisy generators.
4. Pollutants produce harmful effects on the body.

D. Answer the following questions.

1. What is air pollution?

2. How is water polluted? How can we control water pollution?

3. What harm does noise pollution cause?

4. What are the major causes of air pollution? What can we do to control them?

5. List some ways of reducing the noise pollution.

Skill Based Learning

A. Find out the cause and effects of soil pollution in the environment. Write them in the space provided.

Life Skills and Values

B. Combustion of fossil fuels generates a lot of air pollution. Suggest any two alternative sources of energy which do not cause any pollution.

RACK YOUR BRAIN.

Creative Thinking

- Write the changes you would suggest in your home in order to be environment friendly.

TEACHER'S NOTE

Describe the types of air pollution to the students in detail.



LIGHT AND SHADOW

- What Happens When an Object Comes in the Path of Light?
- Solar Eclipse
- Lunar Eclipse

We already know, the Sun, electric bulb and a candle are sources of light. We need light to see things. At night, we cannot see things in a room unless we switch on the bulb. We see an object only when the light that falls on it, bounces back and enters our eye.

The Sun is a source of light. The Sun is millions of kilometres away from us. Yet its light reaches the earth. This shows that light can travel from one place to another. We are able to see objects during the day because of the sunlight.

The Moon does not have its own light, yet we see the Moon shining brightly in the sky, at night. It appears bright because light from the Sun falls on it.



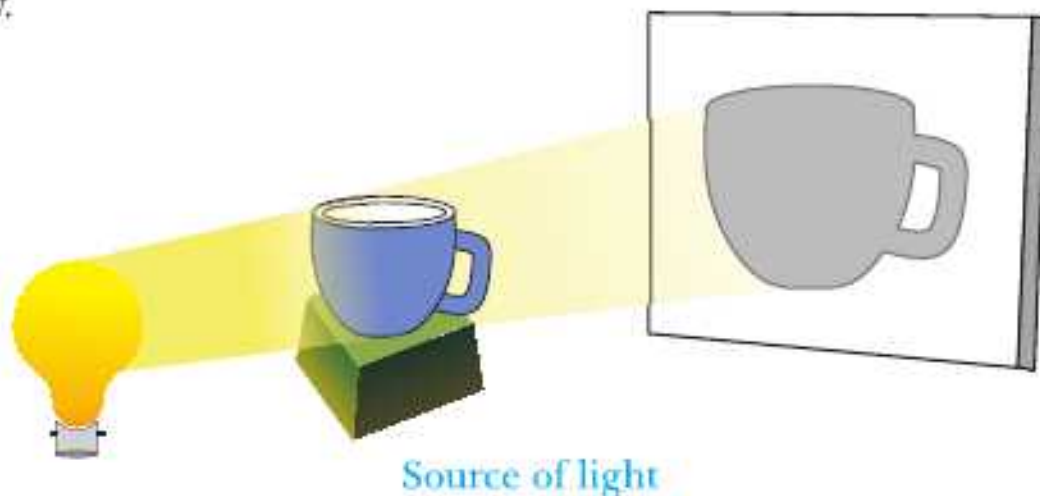
The Moon gets its light from the Sun.

Light travels in a straight line, but when the path of light is blocked by the wooden pieces, we cannot see the entire twig. If the wooden pieces are removed, the entire twig will be visible.

What Happens When an Object Comes in the Path of Light?

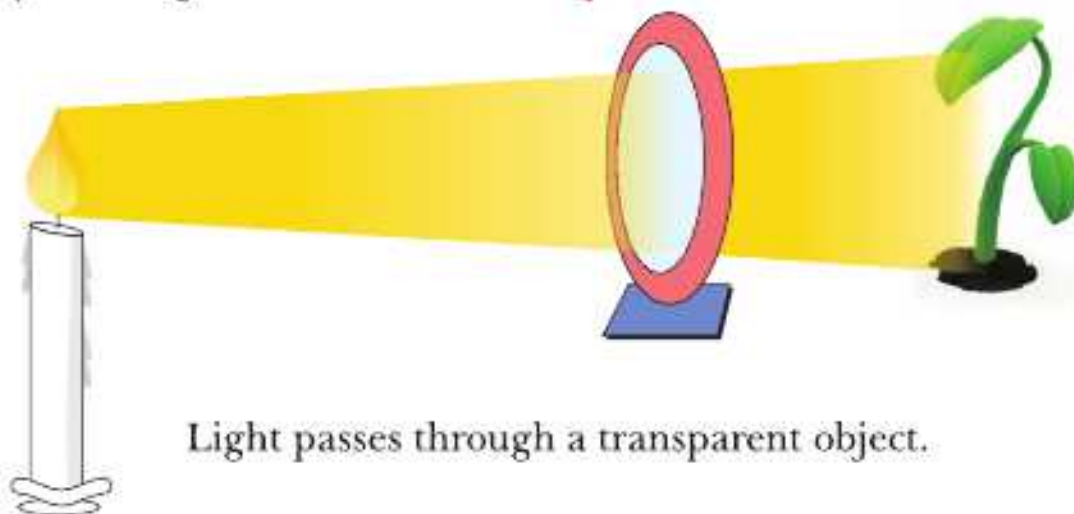
An object which does not let the light pass through, is called an opaque object. We cannot see through an **opaque** object. Wood, stone, and metals are opaque materials. When an opaque object comes in the path of light, light cannot travel further, it gets stopped.

When light is stopped in its path by an object, a dark area is formed on the surface behind the object. This dark area is called a **shadow**. An opaque object forms a clear shadow. The object which forms a shadow is always between the source of light and the shadow.

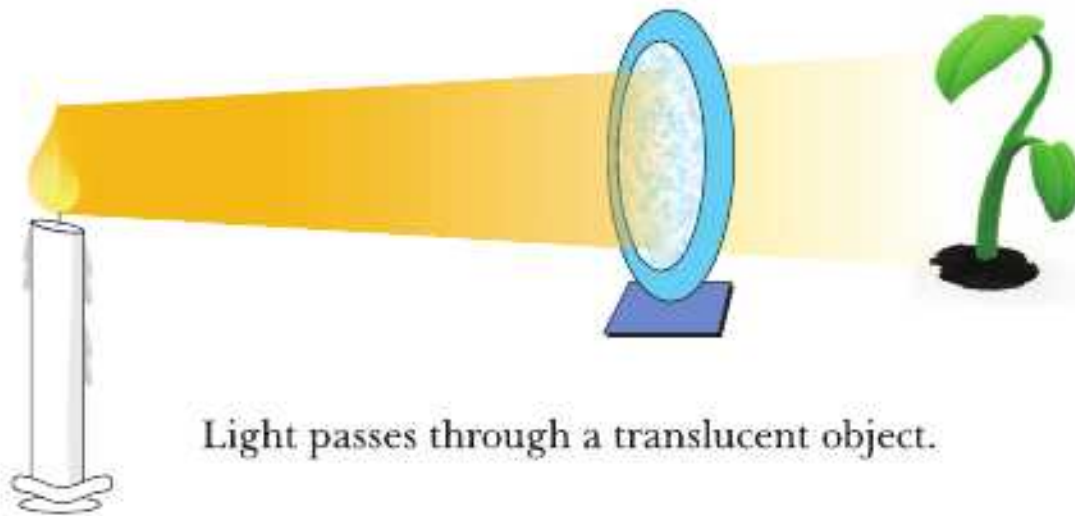


Light does not pass through an opaque object.

Objects made of clear glass and plastic allow the light to pass through them completely. Such objects are said to be **transparent**.



Some objects allow only some amount of light to pass through them. Such objects are called **translucent**. Certain type of plastics and glass are translucent materials. We cannot see through translucent objects clearly. Translucent objects form **faint shadows**.



Light passes through a translucent object.

Solar Eclipse

The Moon moves around the Earth. Once a month, the Moon comes directly between the Earth and the Sun. This phase of the Moon is known as the **new moon**.

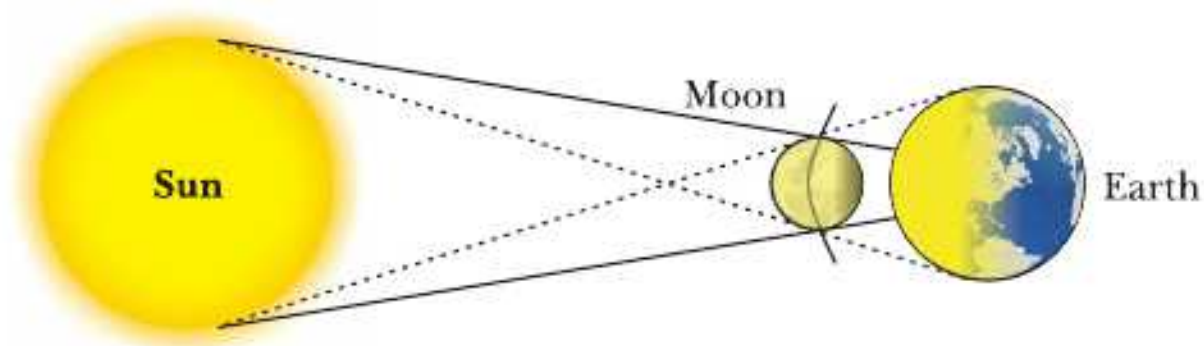
Sometimes on a new moon day, the Moon's shadow falls on certain parts of the earth. From these parts, the sun appears covered by the moon and we see a **solar eclipse**. A solar eclipse occurs during the day on a **new moon day**.

During a solar eclipse, the portion of the Moon that locks the Sun, appears as a dark patch against the brightness of the sun.



Have a Look!

A solar eclipse can be viewed by properly designed and manufactured solar filters. Ordinary sunglasses are not safe for viewing the solar eclipse.



Solar eclipse

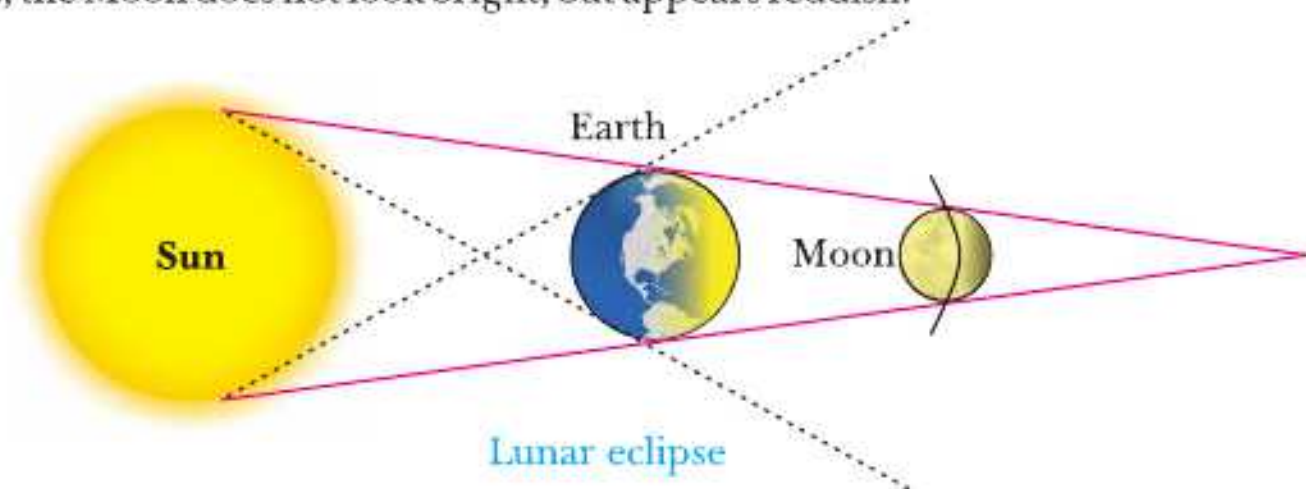
When the Sun gets completely blocked by the Moon, it is total solar eclipse. A total solar eclipse lasts only a few minutes.

Sometimes, the Sun is not completely blocked by the Moon. This is called **partial solar eclipse**. During a partial solar eclipse, a part of the Sun is visible.



Lunar Eclipse

On a **full moon night**, the Earth passes between the Sun and the Moon. On some such nights, the Earth blocks the sunlight falling on the Moon and the Moon gets covered with the Earth's shadow. This causes a **lunar eclipse**. When the lunar eclipse occurs, the Moon does not look bright, but appears reddish.



At times, the Earth partially blocks the sunlight falling on the Moon. This is called **partial lunar eclipse**.

A lunar eclipse lasts for a few hours. Unlike the solar eclipse, a lunar eclipse can be viewed with the naked eye.



- ♦ Light travels in a straight line.
- ♦ When light is stopped by an object, a shadow is formed.
- ♦ A solar eclipse occurs on a new moon day.
- ♦ A lunar eclipse occurs when the Earth blocks the sunlight and the Earth's shadow falls on the Moon's surface.
- ♦ A lunar eclipse occurs on a full moon night.



A. Tick (✓) the correct option.

1. No shadow is formed when the object is:
a. transparent ☐ b. translucent ☐ c. opaque ☐
2. A solar eclipse occurs when the:
a. Earth is between the Sun and the Moon ☐
b. Moon is between the Sun and the Earth ☐ c. Sun becomes black ☐
3. Light always travels in a:
a. straight line ☐ b. wavy line ☐ c. curved line ☐
4. A lunar eclipse occurs when the:
a. Earth is between the Sun and the Moon ☐
b. Moon is between the Earth and the Sun ☐ c. Moon is bright ☐

B. Fill in the blanks.

1. Light travels in a _____ line.
2. A transparent object does not form a _____.
3. The solar eclipse occurs during the day on a _____ day.
4. A lunar eclipse lasts for a _____ hours.
5. The Moon gets its light from the _____.

C. Write short notes on the following.

1. Transparent objects
2. Solar eclipse
3. Partial lunar eclipse
4. Translucent objects
5. Lunar eclipse

D. Answer the following questions.

1. What is the path taken by the light rays?

2. What are opaque objects? Give an example.

3. When does a lunar eclipse occur?

4. What is a transparent object? Give an example.

5. How is an opaque object different from a translucent object?

Skill Based Learning

A. Compare solar and lunar eclipse.

Art Integration

B. Draw a simple labelled diagram to show how a solar eclipse occurs.



Life Skills and Values

C. Consider a wooden stick about half a meter long. Fix one end of it in the open ground where there are no trees and buildings near it. Look at its shadow in the morning and mark its points of shadow. Note the direction of Sun with respect to stick. Where is shadow of the object formed?

RACK YOUR BRAIN.

Creative Thinking

- In a completely dark room, if you hold up a mirror in front of you, will you see a reflection of yourself in the mirror?

TEACHER'S NOTE

Discuss the solar eclipse and lunar eclipse to the students in detail.



OUR WATER RESOURCES

- Water and Its Source
- Importance of Water

Water and Its Source

All living beings need water to live. Water is the most important resource on the earth. There would be no life on earth without water. We need water for drinking, cooking, washing, cleaning and irrigation. Plants and animals also need water.

The main source of water is rain. Farmers depend on the rains for growing crops.

Sometimes, it does not rain in a few parts of our country or a few areas do not receive sufficient rainfall. Irrigation is done to overcome this problem. Our government is trying to reduce our dependence on rainwater.

Irrigation means supplying the water to crops. Three main means of irrigation in India are: canals, wells and tubewells, tanks.

Canals

Water from rivers or dams is diverted to the fields through small channels called **canals**. The canals can take the river water to very distant places. Canal can also be called man-made rivers. **Indira Gandhi Canal** provides water to a large part of Rajasthan. Some of the main canals of our country are:

- Sirhind Canal
- Buckingham Canal
- Agra Canal
- Indira Gandhi Canal



Wells and Tubewells

In some areas where underground water is found easily, wells are dug. A **well** is an open hole dug into the ground to reach the underground water. Well water is used for domestic needs and agriculture.

Sometimes the underground water is very deep. Then a very deep bore is dug into the earth with drilling machine. Tubes are sunk into these bores. Water is brought up with the help of a pump run by electricity. These wells are called **tubewells**. They are very common in Haryana, Punjab and Uttar Pradesh.



Tank

In the Southern plateau, it is difficult to dig land because the land is uneven and rocky. Rainwater is collected in natural hollows (tanks and ponds). Tanks are commonly used in Tamil Nadu, Karnataka and Andhra Pradesh.

Dams

A dam is built across the river to check the flow of water. A huge lake is created behind the dam. There are various channels which control the flow of water. It acts as a **reservoir** to supply water. The stored water is allowed to fall from a height to produce electricity. This type of electricity is called **hydroelectricity**. Some dams serve many purposes and are called multipurpose projects.



They help in:

- Storing water
- Checking floods
- Generation of hydroelectricity
- Providing infrastructure for fishing



Have a Look!

Our first Prime Minister, Jawahar Lal Nehru, called dams the '**Temples of Modern India**'.

Some of the multipurpose projects in India are:

- **Bhakra Nangal Dam** built on Satluj river in Punjab.

- **Damodar Valley Project** on the Damodar river in Jharkhand.
- **Nagarjuna Sagar Dam** on the Krishna river in Andhra Pradesh.
- **Farakka Baragge** on the Ganga river in West Bengal.
- **Hirakund Dam** on the Mahanadi in Odisha.
- **Sardar Sarovar Dam** across the Narmada river in Gujarat.
- **Mettur Dam** on the river Kaveri in Tamil Nadu.

Importance of Water

Water is our basic need. In cities and towns, water is supplied through taps. In villages, people get water through the wells, tanks, rivers, ponds and handpumps.

We cannot use water of sea for drinking or irrigation because it is salty.

We should use clean and pure water for drinking and cooking. If the water is not pure, we must boil it before drinking.

Water is precious. We must not waste it.



- ♦ Rain is the main source of water.
- ♦ We need water for drinking, cooking, washing, cleaning and irrigation.
- ♦ Dams are built across the rivers to regulate the water.
- ♦ Tanks, wells, canals, tubewells are the main methods of irrigation in India.
- ♦ We should not waste water.



A. Tick (✓) the correct option.

- The Mettur Dam is built on the river:

a. Kaveri	☐	b. Krishna	☐	c. Satluj	☐
-----------	--------------------------	------------	--------------------------	-----------	--------------------------
- Tubewells are run by:

a. hand	☐	b. electricity	☐	c. tap	☐
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- Tanks are commonly used in:

a. Kanpur	☐	b. Punjab	☐	c. Tamil Nadu	☐
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B. Fill in the blanks.

1. _____ is the main source of water.
2. A _____ is an open hole dug in the ground.
3. We should always drink _____ water.
4. In cities water is supplied through _____.

C. Write 'T' for true and 'F' for false.

1. Water is not useful for us.
2. Rainwater is one of the sources of water.
3. Tubewells are not multipurpose projects.
4. Dams produce electricity.

D. Answer the following questions.

1. Why do we need water?

2. How is electricity made from water?

3. Why are dams called multipurpose projects?

4. How are canals useful? Name three states where canals are used.

Skill Based Learning

Life Skills and Values

- Providing drinking water to people on a hot summer day is a noble practice. At some places, people keep clean drinking water in surahis or matkas by the roadside as charity. This is known as piao. A piao serves water to thirsty people free of cost.

RACK YOUR BRAIN.

Creative Thinking

- In most of the houses, tap water is passed through a filter before being used for drinking. Why?

TEACHER'S NOTE

Explain the importance of rivers and dams to the students in detail.



OUR ENVIRONMENT

- Global Warming
- Conservation of Wild Animals
- Greenhouse Effect
- Conservation of Forests

Human activities are increasing the greenhouse effect. When people drive a car or operate a factory, they burn coal, oil and other fuels. This adds extra greenhouse gases to the air and the extra gases trap more heat. This has led to **global warming**. Global warming is the steady rise in the average temperature of the Earth's surface.

Global Warming

The warmer weather could harm living things. It could also cause the polar ice caps to melt. This would cause the sea levels to rise. Plants, animals, buildings along coastlines would be in danger.



Use of Three R's a Must in your Daily Life

1. **RECYCLE** - Metals, paper, plastics, etc. can be recycled to make new items.
2. **REUSE** - Books, clothes can be reused by a needy person.
3. **REDUCE** - Buy what you need, don't waste or throw away anything.

Let us all join our hands and save the Earth.

If each one of us does one's bit to conserve electricity, water, fuel, etc. Daily, think, how much cleaner and greener our world will be!

Greenhouse Effect

Have you ever seen a greenhouse?

Greenhouses are used to grow plants in winter. It is a house made of green plastic or glass which traps the heat of the Sun to allow plants to grow in winter.

Similarly, some gases trap the heat of the Sun in the atmosphere of our planet to help us survive. These heat trapping gases are called greenhouse gases. The most common greenhouse gases are water vapour, carbon dioxide, and methane. Without the greenhouse effect, the Earth would be too cold for life to exist.

Land, oceans and plants absorb or soak up energy from sunlight. They release some of this energy as heat. Greenhouse gases absorb the heat and then send it back towards the Earth. Without greenhouse gases, this heat would escape back into space.



Conservation of Wild Animals

Cutting and burning of the trees have forced the wild animals to flee from their natural habitat. Many animals are dying or being killed. Some animals are being killed for their medicinal value.

Some animals are already extinct. Some of them are considered endangered.

Tigers and rhinoceroses in India are getting less every year due to habitat loss and poaching. The great Indian bustard, a beautiful bird nearly the size of a young ostrich, is almost extinct.





Conservation of Forest

Deforestation is the major cause of the 'greenhouse effect' as fewer trees use carbon dioxide when they make their own food. Fewer trees mean that less carbon dioxide is being taken out of the atmosphere. More the trees, less would be the pollution; lesser the trees, more would be the pollution.

Forest also gives shelter and protection to many animals. Due to deforestation, some animals are dying as there is a decline in their prey and animals do not get enough to eat.

Trees give us life-giving oxygen, food, and medicines.

Planting and looking after trees is very essential. This is called **afforestation**.



Have a Look!

National park and wildlife sanctuary are places where wild animals are protected in their natural habitat.

Can you imagine the world without plants and animals? How will we balance the food chain?



- ♦ Carbon dioxide, methane, and water vapour are the main greenhouse gases.
- ♦ The trend of steady increase in the temperature of the Earth's surface is called global warming.
- ♦ We should plant more trees and conserve energy wherever possible.
- ♦ Tigers and rhinoceroses are considered endangered animals.



A. Tick (✓) the correct option.

- This animal is almost extinct:
a. Cow ☐ b. Young ostrich ☐ c. Tiger ☐
- _____ is a greenhouse gas.
a. Nitrogen ☐ b. Oxygen ☐ c. Sodium hydroxide ☐
- Planting and looking after trees is called:
a. global warming ☐ b. deforestation ☐ c. afforestation ☐

B. Fill in the blanks.

- Global warming is _____ in temperature of the Earth's surface.
- The Great Indian bustard is an _____ animal.
- Melting ice caps in the _____ regions may lead to increase in the sea level.

C. Write 'T' for true and 'F' for false.

- Oxygen is a greenhouse gas.
- Cow is an endangered animal.
- Trees use carbon dioxide to make their food.
- Deforestation is a major cause of global warming.

D. Answer the following questions.

- How does global warming affect us? Give two reasons.

- Why is it important to protect forests?

- What are the three R's that we need to use in our daily life?

- Explain the 'greenhouse effect'.

5. Give five points as to how you can help to save the Earth?
-

Skill Based Learning

- A. Write down various means to improve environmental problems faced by the world today and about 25 years back.

Life Skills and Values

- B. Listing a greenhouse gas, write its possible source of production on a larger scale. What are its harmful effects?

RACK YOUR BRAIN.○

Creative Thinking

- What will happen if there are too much greenhouse gases in the atmosphere.

TEACHER'S NOTE

Explain the importance of forests to the students in detail.



Our Universe

THE EARTH, SUN AND MOON

- The Earth
- The Sun
- The Moon
- Phases of the Moon
- Observation and Exploration

The planet Earth on which we live, is a huge ball revolving around the Sun. The Earth revolves at 1,08,000 km per hour.

Scientists believe that millions of years ago, our Earth was just a huge cloud of dust and gases floating in space. As the Earth spins it began to cool first and formed a crust. But the inside of the Earth is still bubbling hot-molten! Even today our planet is in the process of cooling.

The Earth is not a perfect sphere; it is slightly flattened at the poles and bulges at the equator. It has a diameter of 12,000 km at the equator. Scientists believe that the Earth is about 5,500,000 years old. It is about 150 million km away from the Sun.

The Earth

The structure can be divided into three parts :

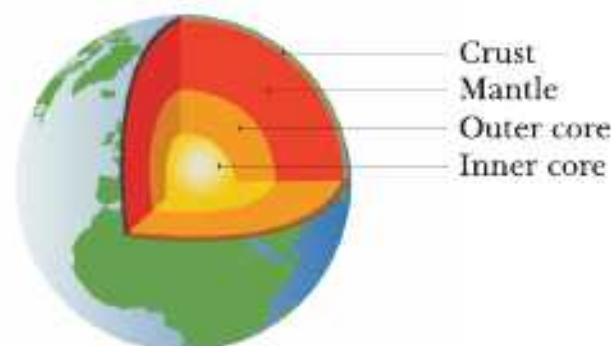
1. The crust
2. The mantle
3. The core

The Crust

It is the outermost layer of the Earth on which we live. Plants, animals and people live on the crust. It is the thinnest layer.

The Mantle

Rocky mantle lies beneath the crust. The upper part of the mantle is hard and it merges with the crust. The lower part of the mantle is hot and molten like lava.



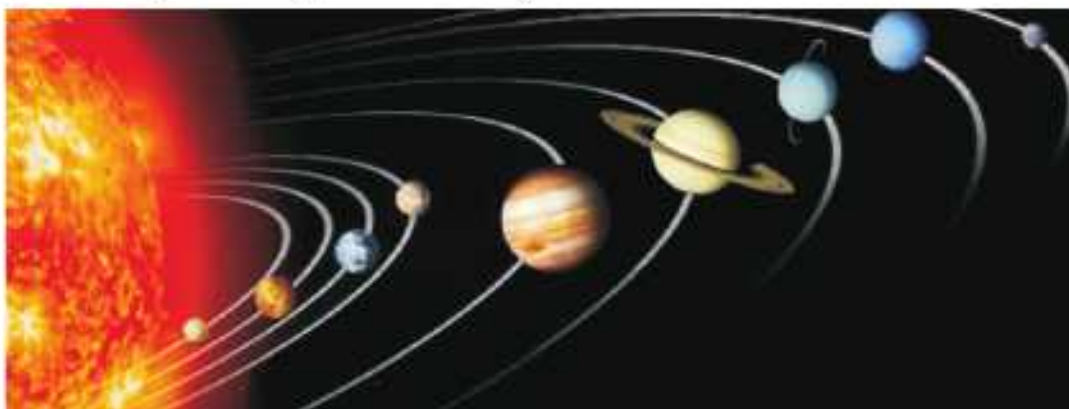
The Core

The core is the center of the planet. It is about 6,900 km across. The core is very hot. It has two parts: the outer core and the inner core. The outer core is in the liquid state which contains iron and nickel. The inner core is like a solid ball made of iron.

The Sun

The Sun is the star at the centre of the solar system. It is a hot ball of gases that gives off great amounts of energy. Life on Earth depends on light and heat from the Sun.

The solar system also contains planets, comets, and asteroids. They all orbit or travel around the Sun. Earth orbits the Sun at a distance of about 15,00,00,000 km. The next closest star to Earth is about 250,000 times farther away. That is why the Sun is by far the brightest object in the sky.



The Sun consists mostly of the gases hydrogen and helium. The Sun has been shining for millions of years. Billions of years in the future, the Sun will use up all its hydrogen fuel. Eventually it will stop shining.



Have a Look!

The Sun is about 400 times larger than the Moon yet when seen from the Earth, both look the same size because the Sun is about 400 times farther away.

The Moon

The Moon is a large natural object that travels around the Earth. After the Sun, it is the brightest object in the sky.

The Moon is slightly more than one quarter the size of the Earth. Its diameter is about 3,500 km.

There is no atmosphere on the Moon. So, it is very hot where the sunshine falls.



The Moon's gravity is one-sixth of the Earth's gravity. You can lift a heavy object easily on the Moon. Moon's gravitational pull is $\frac{1}{6}$ that of Earth. If your weight is 60 kg on the Earth, it will be 10 kg on the Moon.

The Moon spins about its centre. It completes one rotation in about 27 days, the same time it takes to complete one orbit. For this reason, the same side of the Moon always faces the Earth.

Phases of the Moon

When viewed from the Earth, the Moon looks different at different times. These varying appearances are called phases. Sometimes the Moon looks like a full circle. At other times, it appears as only a thin slice or looks completely dark. However, the Moon's shape does not change, it just looks that way from the Earth. The Moon reflects light from the Sun. As the Moon orbits the Earth, the Sun shines on different parts of the Moon. This causes different parts to be visible from the Earth.



Observation and Exploration

People have observed the Moon since ancient times. In the 1600s, the invention of the telescope allowed the people to study the Moon more closely.



21st July, 1969 was a historical day for mankind because the American spacecraft Apollo-11 landed on the moon on this day.

Neil Armstrong was the first man to step on the Moon followed by Edwin Aldrin. Mike Collins remained in the spacecraft which orbited the Moon. They collected samples of Moon rock and dust to study more about the Moon.

- ♦ The lower part of mantle is hot and molten.
- ♦ The core is the centre of the planet.
- ♦ Life on earth depends on light and heat from the Sun.
- ♦ The Moon has two types of movement : orbit and spin.
- ♦ There is no atmosphere on the Moon.



A. Circle the odd one out.

crust	Moon	core	mantle
Mercury	Venus	Earth	Crust
Sun	Moon	Earth	satellite

B. Fill in the blanks.

1. The _____ layer of mantle is in a molten state.
2. The Moon's gravity is _____ than the Earth's gravity.
3. _____ was the first man to step on the Moon.
4. Earth orbits the Sun at a distance of about _____ kilometres.
5. The Sun is the star at the centre of the _____.

C. Correct the statements.

1. Mike Collins was the first man to step on the Moon.
2. The mantle is the thinnest layer of the Earth.
3. The gravity of the Moon is 10 times less than the Earth's gravity.
4. The structure can be divided into four parts.
5. The core is the centre of the Sun.

D. Answer the following questions.

1. Describe the structure of the Earth.

2. Describe the Moon's surface.

3. What effect does the Moon's gravity have on the Earth?

4. What are the phases of the Moon?

Skill Based Learning

A. Find out the names of planets that have life on them.

B. Is the Sun hotter on the surface or the centre?

Life Skills and Values

C. Most people blame stars and planets in their horoscope for being unsuccessful in their lives. Ajay is one of them. He believes that positioning of Saturn in wrong place in his horoscope is the reason of his bad performance in studies.

- What are the heavenly bodies?
- Do you really think that these heavenly bodies play an important role in our failure or success?
- Have you ever heard this kind of things regarding your failure, success or before doing any auspicious work? From whom?
- What value of Ajay is shown here?

RACK YOUR BRAIN.

Creative Thinking

- Assume, the distance between Earth and Sun becomes half of its present distance. What is likely to happen to life?

TEACHER'S NOTE

Explain the solar system to the students in detail.



Elements and Metals

ELEMENTS AND COMPOUNDS

- Atoms
- Molecules
- Elements
- States of Matter

Anything that has mass and occupies space is matter.

All substances are formed of extremely small particles. A tiny portion of any material will have millions of such particles which in turn are made up of atoms.

Not only sugar, but all kinds of substances are formed of extremely small particles called atoms. A tiny portion of any material will have millions of atoms.



A sugar crystal.



Made up of many particles of sugar.



Each sugar particle is made up of extremely small particles called atoms.

Atoms

Atoms are so small that thousands of them can fit on the tip of a pencil point. Not only man-made things like a needle or a car, but even the dust floating in the air, dew drops that settle on the grass, blue whales of the ocean, germs that cause diseases, and the mighty Sun are made up of atoms.

Elements

A substance that is made up of a single type of atom, is called an **element**. Oxygen, chlorine, gold, hydrogen and carbon are examples of elements. Some elements, like iron, gold, silver, and copper are metals while some, like carbon, oxygen and nitrogen are not metals.

Atoms found in one element are not found in any other element. Every kind of atom has its own unique properties. That is why every element is different from all other elements. For example, silver element is different from other elements.



Have a Look!

The atoms of a single kind are coloured alike and different atoms are coloured differently.

Each element has a specific name. The **International Union of Pure and Applied Chemistry** (IUPAC) approves names of elements. Since there are large number of elements, it is easy to refer to them by symbols rather than use their full names all the time.

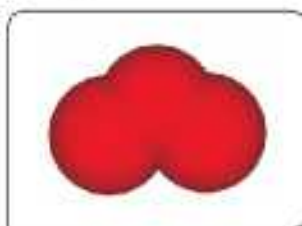
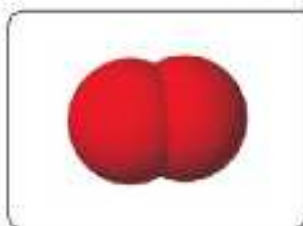
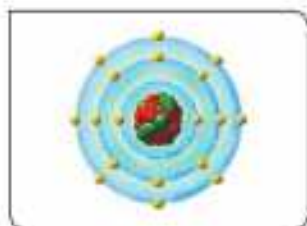
Element	Symbols	Element	Symbols
Alluminium	Al	Hydrogen	H
Bromine	Br	Iodine	I
Chlorine	Cl	Potassium	K
Cobalt	Co	Magnesium	Mg
Copper	Cu	Sodium	Na
Iron	Fe	Sulphur	S
Helium	He		

Most elements are obtained from nature. There are 92 naturally occurring elements. Only a few can be made in the laboratory by scientists. Most man-made elements can only be made in very small quantities.

Molecules

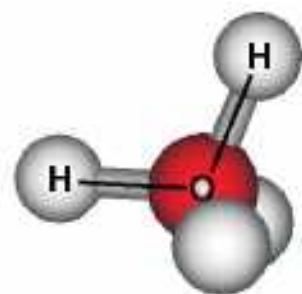
The smallest part of an element that has all its properties is its **molecule**. A molecule of an element is made up of one or more atoms of the same kind.

- A molecule of argon is made up of 1 argon atom.
- A molecule of oxygen is made up of 2 oxygen atoms.
- A molecule of ozone is made up of 3 ozone atoms.
- A molecule of sulphur is made up of 8 sulphur atoms.



Compounds

The molecules of some substances are made up of more than one type of atoms, such substances are called **compounds**. Water is a compound. The water molecule is made up of two different types of elements—hydrogen and oxygen. Two hydrogen atoms and one oxygen atom join together to form a molecule of water. The molecule is the smallest part. It has all the properties of a compound. A molecule of water has all the properties of water. Carbon dioxide, table salt and sugar are examples of some other compounds.



States of Matter

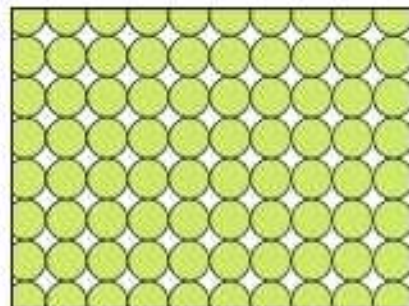
Molecules and the Three States of Matter

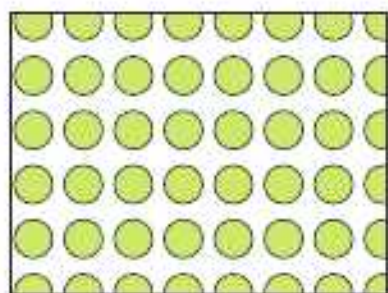
At room temperature, sugar is a solid, while water is a liquid, and the carbon dioxide that we breathe out is a gas. Every substance is either a solid or a liquid, or a gas. Solids, liquids and gases are the three states of matter.

What Decides the State of the Substance?

You have learnt that every substance is made up of molecules. It is the **arrangement of molecules** that decides the state of the substance.

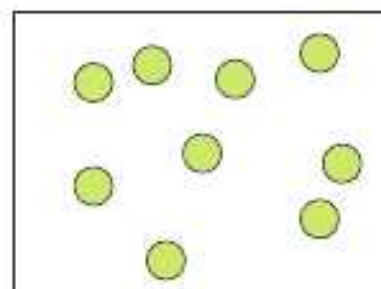
- When the molecules are closely packed together, the substance is in the solid state.





- When the distance between the molecules **increases**, the substance melts and changes into a liquid.

- When the molecules are far away from each other they are in the gaseous state.



- An atom is the smallest part of matter.
- The distance between the molecules of a substance makes it a solid, liquid or gas.
- Elements and compounds are a part of our everyday life.



A. Tick (✓) the correct option.

1. A substance that is made up of a single type of atom, is called an/a:
 a. symbol ☐ b. element ☐ c. compound ☐
2. A molecule of ozone is made up of _____ ozone atoms.
 a. 2 ☐ b. 1 ☐ c. 3 ☐
3. When the molecules are far away from each other, they are in the _____ state.
 a. gaseous ☐ b. liquid ☐ c. solid ☐

B. Fill in the blanks.

1. A _____ portion of any material will have millions of atoms.
2. Every kind of atom has its own unique _____.
3. Most elements are obtained from _____.
4. At room temperature, sugar is _____.

C. Write 'T' for true and 'F' for false.

1. A molecule is always made up of only two atoms.
2. Bones, muscles, blood and everything else in the human body is made up of atoms.
3. A compound has the properties of the elements that make it.
4. A compound is also made up of just one type of atom.

D. Classify each substance as a 'compound' or an 'element'.

Element	Compound

iron
carbon dioxide
table salt
oxygen
gold
sugar
silver
water

E. Answer the following questions.

1. What is a compound?

2. Why is water a compound?

3. What are the three states of matter?

4. Describe the molecule of oxygen.

5. Name the molecule of two atoms.

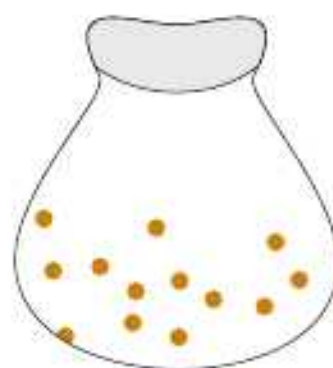
Skill Based Learning



solid



liquid



gas

A. Observe the pictures given above carefully. Answer the questions. The blue dots represent the molecules of a substance.

1. In which state are the molecules far apart from each other?

2. In which state are the molecules arranged very close to each other?

3. Does the distance between the molecules decide the state of a substance?

Life Skills and Values

B. Classify the following into elements and compounds:

Bromine, Water, helium, Sugar, Carbon

RACK YOUR BRAIN.

Creative Thinking

- Give the names of 6 common metals along with their symbols.

TEACHER'S NOTE

Explain the three states of matter to the students in detail.



METALS AND MATERIALS

- Sources of Metals and Materials
- Man-made Materials

- Materials from Plants and Animals

We use several things in our daily life. All these things are different in shape and size from each other and made of different metals and materials. All these metals and materials have different properties, so the things made from them are also different.

Tin, iron and gold, etc. are some metals and wood, paper, plastic, glass and rubber are some materials that are used to make different things.

Sources of Metals and Materials

We get metals and materials from nature. Some are obtained from the earth and some are obtained from the plants and animals.

We get different metals from the earth. Metals are used for making machines, utensils, vehicles, electric appliances, wires and many other things. Iron, aluminium, copper, etc. are some common metals. Gold and silver are the metals used for making jewellery.

Sand is used for making glass. Clay is used for making earthen pots.

The crude oil which we dig out from the earth gives petrol and diesel, which are used to run machines and vehicles.

We also get kerosene, wax, vaseline, etc. from petroleum, that are used in industries.

Materials from Plants and Animals

Timber

We get timber from trees. Trees like **sal**, **teak** give us wood which is very hard and strong. This wood is used to make furniture, houses, ships and many other things.



Rubber

Rubber is made from the liquid obtained from the trunk of the rubber tree. Rubber is highly stretchable.

Paper

Paper is made from the wood of the bamboo and eucalyptus tree. The wood of these trees is soft of which a watery paste is made. This paste is then used to make paper.



Fibre

We get cotton from the cotton plants and jute from the jute plants. The fibre of flax is used to make linen. Cotton fibre is used to make cloth while rugs and ropes are made from the jute fibres.

Leather, Silk and Wool

We get leather from the skin of the dead animals. It is used to make leather goods. Silk thread is obtained from the cocoon of the silkworm. We get wool from the hair of the sheep, camels, etc. All these materials are used to make clothes.



Man-Made Materials



Plastic

Today plastic has become as important as wood and metals. We make many things like **ropes**, **toys**, **baskets**, etc. from plastic. Plastic is made from different materials.

Synthetic Fibre

Nylon, **rayon** and **polyester** are some synthetic fibres that are made in factories. They are used to make **wind cheaters**, **raincoats**, **umbrellas**, etc. They are also used to make **synthetic** clothes.



Have a Look!

The Earth is five billion years old, although life has existed 150 million years ago.

Things made of plastic do not get decomposed like the things made of natural materials. They last for a long time. This is a useful property but the other side of the picture is that plastic stays in soil forever and pollute it. Thus, we should avoid the use of plastic things to keep our environment clean.



Have a Look!

- There are three states of matter— solid, liquid and gas.
- In many countries and in many parts of our country also, plastic bags are banned by the government.



Recollect

- ♦ We get different materials from the nature for making different things.
- ♦ Wood, paper and fibre come from the plants. We get fibres like wool, silk and fur from animals.
- ♦ Metals and petroleum come from the earth's crust.



Let's Do!



A. Tick (✓) the correct option.

1. Wood is used for making:
a. utensils ☐ b. furniture ☐ c. machines ☐
2. One of these is a synthetic fibre:
a. Rayon ☐ b. Cotton ☐ c. Flax ☐
3. Rubber is obtained from:
a. plant ☐ b. petroleum ☐ c. animals ☐

B. Fill in the blanks.

1. _____ is used for making glass.
2. _____ is highly stretchable.
3. We get _____ from the skin of dead animals.

C. Write 'T' for true and 'F' for false.

1. Paper is made from the bamboo tree.
2. Plastic is a natural material that we get from the earth's crust.
3. Plastic cannot be decomposed easily.

D. From where do we get these materials.

1. Cotton _____
2. Glass _____
3. Petrol _____
4. Plastic _____
5. Polyester _____

E. Answer the following questions.

1. Mention the sources of metals.

2. What is the use of rubber in our life?

3. What do you know about the fibres?

4. Discuss the man-made materials.

Skill Based Learning



Experimental Learning

- A. Make a list of the natural and man-made things in your house.

Natural things	Man-made things
_____	_____
_____	_____
_____	_____
_____	_____
_____	_____

- B. You all would have visited the petrol pumps. This time when you go to the petrol pump, observe the place carefully. Write five lines on the activities going on in a petrol pump.

Life Skills and Values

- C. Plastic is a threat to the environment. We should stop using things made of plastic.
- Stop using polybags.
 - Start carrying jute/paper bags to the market.
 - Do not throw the things made of plastic in the open as they pollute your surroundings.

RACK YOUR BRAIN.○

Creative Thinking

- What are some properties of your pencil, a sheet of paper or a metal fork? List the properties of each.

TEACHER'S NOTE

Explain the properties of metals and materials to the students in detail.